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Scheduled service network redesign in freight transportation: Joint demand and service selection under disruption

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ABSTRACT

Natural disasters, including earthquakes, floods, hurricanes, and landslides, have profound impacts on transportation service networks, particularly disrupting rail and road transportation. Such events often damage critical infrastructure, such as terminals and their connecting bridges, tunnels, and railways, leading to prolonged service interruptions. To address these challenges, this paper introduces an innovative extension of the classical Scheduled Service Network Design (SSND) problem, termed the Scheduled Service Network Redesign (SSNR) problem. The SSNR is formulated as a mixed-integer programming (MIP) model that seeks to maximize total profit by balancing the revenue from selected demands against the costs of serving them under disrupted network conditions. The SSNR focuses on reconfiguring the post-disaster network by optimizing demand selection and assigning demands to available services under disruption conditions. A key distinction between SSND and SSNR is that they operate on two different physical network states for the definition of demands and services versus the execution of planning decisions. Although both demands and services are originally defined on the pre-disaster physical network, all subsequent decisions, including demand selection, service activation, and the construction of demand itineraries, must be carried out on the modified network after disruptions occur. This significantly increases the problem complexity, as some demands may become infeasible due to damaged terminals, rail links, or roads, while others may require rerouting through alternative itineraries that can lead to substantially higher transportation costs. To tackle these complexities, this paper proposes a Benders Decomposition (BD) reformulation to divide the SSNR into two subproblems. The efficiency of the BD reformulation is evaluated against an exact solver to demonstrate its performance. In addition, sensitivity analyses are conducted to assess the robustness of the SSNR model. The results provide actionable managerial insights for strengthening the resilience of transportation service networks in the face of natural disasters. These findings serve as a foundation for developing decision support systems, highlighting the effectiveness of the BD reformulation, and offering practical guidelines for improving scheduled service networks and ensuring continuity under disruptive events.

1. Introduction

Natural disasters, including earthquakes, floods, hurricanes, and landslides, pose significant threats to transportation service networks, particularly those relying on rail and road infrastructure (Kulkarni et al., 2022). These networks are critical to economic stability and societal well-being, facilitating the efficient movement of goods and passengers (Li et al., 2024). When disasters strike,

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they may damage or temporarily disable specific infrastructure components, such as terminals, bridges, tunnels, or railway tracks, leading to localized service disruptions, economic losses, and compromised supply chain operations (Scaparra and Church, 2008; Moosavi et al., 2022). For instance, the 2023 Lahaina wildfires in Hawaii caused severe disruptions to key transportation corridors, resulting in localized blockages that delayed emergency response and supply distribution (Roy et al., 2024). Such events highlight the urgent need for transportation service networks that can be reconfigured and operated effectively under post-disaster conditions.

Building resilience in transportation networks offers substantial benefits, including reduced recovery times, mitigated economic losses, and improved emergency response effectiveness (Liberatore et al., 2014). In practice, post-disaster transportation systems rarely experience a complete network shutdown; instead, they typically operate in a degraded state, where a subset of terminals and transportation services becomes partially or fully unavailable while the remaining network continues to function (Liu et al., 2024). Ensuring resilience under such localized disruptions requires optimization frameworks capable of adapting service configurations and reallocating resources in response to damaged network components (Chang et al., 2024). However, classical Scheduled Service Network Design (SSND) models (Taherkhani et al., 2022) generally assume fully operational infrastructure and therefore lack the flexibility needed to capture post-disaster operational realities.

Despite the importance of this issue, developing effective redesign strategies for disrupted transportation networks remains challenging. First, following a disruptive event, it is necessary to rapidly identify the affected demands and infrastructure components and redesign service networks accordingly. Second, the interdependent nature of terminals and transportation services complicates the reconfiguration of demand itineraries when certain nodes or arcs are unavailable. Finally, maintaining profitability while ensuring service continuity under disrupted network conditions constitutes a complex optimization challenge that cannot be adequately addressed by traditional SSND models.

To overcome these challenges, this paper introduces an extension of the classical SSND model, termed the Scheduled Service Network Redesign (SSNR) problem, which is specifically developed for transportation networks operating under localized post-disaster disruptions. In this study, disruptions are modeled as the unavailability of selected terminals (nodes) and transportation services (arcs) in the physical network, while the remaining network remains operational. Although the demands and services are defined based on the pre-disruption physical network, the decisions regarding demand selection and service activation are made under disrupted network conditions. In other words, the demands and services are initially established without considering potential disruptions. In such a context, a major challenge arises because the optimal itinerary associated with a demand in the original disruption-free network may no longer exist after a disruption occurs, while alternative itineraries may become operationally inefficient or economically suboptimal. Consequently, the SSNR problem aims to identify feasible demands and allocate them to the available services in order to construct optimal post-disruption demand itineraries that maximize profitability while maintaining service continuity.

The contributions of this paper are twofold. First, the SSNR problem is formulated as a mixed-integer programming (MIP) model that maximizes total profit by balancing demand revenue against service costs under disrupted network conditions. Second, to address the computational complexity of the proposed model, a Benders Decomposition (BD) reformulation is developed, decomposing the problem into two subproblems to enhance tractability, demonstrating its effectiveness and computational efficiency in large-scale post-disaster network redesign scenarios.

The rest of this paper is organized as follows. Section 2 presents a comprehensive literature review, highlighting prior studies related to SSND problems. Section 3 provides a detailed discussion of the problem definition, focusing on the physical network, demand characteristics, scheduled services, and planning levels. In Section 4, the MIP formulation for the SSNR problem is introduced. Section 5 develops the BD reformulation and its iterative algorithm for solving the proposed SSNR problem. Section 6 reports the experimental results and sensitivity analyses conducted on various test instances. Finally, Section 7 concludes the paper by summarizing the key findings, providing actionable managerial insights, and suggesting potential directions for future research on the network redesign in freight transportation.

2. Literature review

To systematically review the literature and emphasize the unique contributions of the proposed SSNR framework, this section is organized into three subsections. Section 2.1 reviews the existing and relevant studies on SSND. Section 2.2 examines the literature on disruption management and its applications in transportation and logistics networks. Finally, Section 2.3 identifies the key research gaps and highlights the distinctive contributions of this study within the SSND and disruption management fields.

2.1. Literature review on SSND

Since the foundational work on SSND, Crainic (2000) established that SSND problems can be classified as either static or time-dependent, with the latter addressing variations in demand and changes in problem parameters over the schedule length. In an early exploration of SSND problems, Kim et al. (1999) tackled issues related to terminal and arc capacity limits, time window constraints, and restricted service availability within a multimodal express delivery context. Their work emphasized the practical implications of these constraints in real-world logistics.

Further advancing SSND research, Crainic et al. (2016) introduced an extended model that accounts for resource requirements at facilities, enabling more comprehensive and efficient SSND solutions. Expanding the scope of SSND, Crainic et al. (2018) established a link between high-level resource acquisition and allocation decisions and the tactical choices within SSND, incorporating multiple resource types and exploring outsourcing options. This work paved the way for addressing resource allocation challenges in complex transportation systems.

In enhancing SSND reliability for both strategic and tactical decision-making, Hewitt et al. (2019) made significant contributions, particularly by introducing mechanisms to address uncertainties and improve the robustness of the solutions. In the context of integrated transportation systems, Wang and Qi (2019) proposed a two-stage robust optimization model for SSND, incorporating a probability-free uncertainty set to address demand uncertainty. Their robust model effectively balances operational efficiency with flexibility under uncertain conditions.

Focusing on urban logistics, Fontaine et al. (2021) developed a two-tier multimodal city logistics model that incorporates capacity constraints and demand availability. Their approach provides insights into urban delivery optimization under resource limitations. Similarly, Dai et al. (2021) examined China's rail and air transportation integration, proposing a model for multiple terminal allocations that considers travel demand uncertainties. Their work highlights the importance of intermodal solutions in addressing complex logistics challenges.

From recent studies, Taherkhani et al. (2022) addressed tactical planning within an integrated, multistakeholder framework by incorporating revenue management principles into SSND. Their model strategically selects high-value demands and optimizes service bundling to enhance profitability. Meanwhile, Giusti et al. (2023) explored multicommodity multi-service transshipment within the SSND context, demonstrating the advantages of flexible logistics operations and assessing their impact on cost and decision-making processes.

In the realm of urban logistics, Wu et al. (2023) introduced a novel SSND problem tailored for same-day delivery services, addressing capacity limitations at terminals by restricting the number of vehicles loading or unloading simultaneously. To address this issue, they proposed an integer-based heuristic program. Expanding on stochastic considerations, Liu et al. (2023a) proposed an SSND problem with stochastic demands and fixed routes, employing a two-stage stochastic MIP model. Their work designs fixed routes in the first stage and optimizes flexible vehicle routes in the second stage after demand realization. In a related study, Liu et al. (2023b) modeled a trip design problem as an SSND with consistent multiple trips, showcasing the importance of synchronization in service scheduling.

Recently, Shu et al. (2024b) introduced the continuous-time SSND problem, focusing on minimizing operational costs by optimizing transportation schedules and shipment routes over a continuous planning horizon. Their model is unique in incorporating holding costs, which add complexity but provide a more accurate representation of operational realities. Diao et al. (2024) proposed an electric vehicle-based express SSND, considering the impact of limited recharging resources and the power management requirements for each electric vehicle during its entire operational cycle. They developed a MIP model that incorporates a Dantzig-Wolfe reformulation, which enables efficient handling of the problem's complexity. Belieres and Hewitt (2024) developed a stochastic model for Less-than-Truckload (LTL) freight transportation, emphasizing the importance of flexible scheduling mechanisms to help carriers cope with uncertainties in shipment sizes.

More recently, Shu et al. (2024a) proposed a dynamic discretization discovery method for the continuous-time SSND, building on their earlier work (Shu et al., 2024b). Their model aims to minimize total operational costs through the joint optimization of transportation schedules and demand itineraries. Mirzaei et al. (2025) developed a Benders Decomposition (BD) reformulation algorithm to solve a reliable SSND problem under demand uncertainty. Finally, Vaziri et al. (2025) introduced a stochastic SSND model for multicommodity scenarios, aiming to design a resilient service network under uncertain budgets and fluctuating demand conditions.

2.2. Literature review on disruption management

The literature on disruption management has largely evolved within the domain of humanitarian logistics and relief supply chains (Belieres and Hewitt, 2025). This stream of research primarily addresses decision-making under extreme uncertainty and severely disrupted transportation and service infrastructures, with the overarching objective of ensuring the rapid and reliable delivery of essential aid to affected populations (Rodríguez-Espíndola et al., 2025). Within this context, humanitarian logistics encompasses a broad set of strategic and operational decisions, including the location and protection of temporary and permanent facilities, allocation of scarce transportation resources, routing and flow of relief commodities, coordination among heterogeneous carriers, and the integration of multiple transportation modes across multi-tier relief networks (Jin et al., 2025). These decisions are typically made in highly constrained environments characterized by damaged or partially operational infrastructure, congestion, limited vehicle availability, uncertain and dynamically evolving demand, and the need for close collaboration among multiple stakeholders (Leiras et al., 2014).

A prominent research direction within this literature focuses on the protection and disruption of public facilities and services. Early work by Church and Scaparra (2007) introduced the facility protection problem for public services, enabling governments to allocate limited protection resources to safeguard critical facilities against potential disruptions. Building on this foundation, Aksen et al. (2010) formulated a bi-level attacker-defender model that explicitly incorporates budget constraints on the attacker, capturing the strategic interaction between facility protection and interdiction decisions in humanitarian contexts.

Subsequent studies extended this modeling framework to account for uncertainty and more realistic disruption mechanisms. Liberatore et al. (2011) developed a stochastic optimization model for facility protection that considers uncertainty in the number and realization of disaster scenarios. Aksen and Aras (2012) further investigated a bi-level facility location problem in which a defender opens facilities to serve customers based on proximity while accounting for potential disruptions to unprotected facilities. This line of research was extended by Aksen et al. (2014), who introduced partial interdiction, allowing attackers to destroy only a portion of a facility's capacity, and incorporated demand outsourcing to serve customers who cannot be assigned to any operational facility.

The facility protection framework for public services was later embedded into a more complex network for hierarchical facilities. Aliakbarian et al. (2015) extended the public service protection model to a multi-tier hierarchical facility location problem, where demand satisfaction requires flows through multiple interdependent facilities. To solve the resulting bi-level model, they proposed a hybrid metaheuristic approach combining Simulated Annealing (SA) and Variable Neighborhood Search (VNS). Similarly, Fathollahi-Fard and Hajiaghaei-Keshteli (2018) introduced a bi-objective bi-level model that jointly considers facility protection using multiple defensive systems and capacity expansion under imminent disaster scenarios, aiming to balance protection budgets and total service network costs.

Some studies have also examined disruption management in specialized humanitarian and medical supply chains. Khanduzi and Sangaiah (2019) proposed a bi-level model for a biomedical supply chain network incorporating capacity and budget constraints induced by protection and interdiction decisions. Recognizing the NP-hardness of the problem, they developed a hybrid solution approach combining a Genetic Algorithm (GA) for the upper-level problem with a fast branch-and-cut algorithm for the lower-level problem, demonstrating superior performance compared to a GA-CPLEX benchmark. In a related vein, Forghani et al. (2020) formulated a bi-level humanitarian logistics model aimed at preserving customer satisfaction under malicious attacks on capacitated hierarchical facilities and solved it using a tailored heuristic combining constructive and iterative improvement procedures.

More recent studies have further advanced the modeling of disruption management by considering multiple disaster scenarios, stochastic recovery, and interdependent infrastructures. Li et al. (2022) developed a bi-level facility location and protection model in supply networks facing multiple non-cooperative attackers, explicitly modeling imperfect attack and protection effects. Using a two-stage stochastic programming framework, they optimized post-disruption demand allocation and proposed a hybrid solution method combining GA and column generation. Their results highlighted the significant impacts of limited choice and risk propagation on optimal protection strategies. Most recently, Kuttler et al. (2024) proposed a tri-level protection-interdiction-restoration model for multi-commodity, interdependent critical infrastructure systems, capturing heterogeneous commodity priorities and recovery dynamics. They employed a combination of Benders' decomposition and set-covering decomposition to efficiently solve the resulting large-scale problem. Last but not least, Jin et al. (2025) introduced a last-mile humanitarian logistics planning problem supported by unmanned aerial vehicles. Their framework simultaneously addresses decision-making in both pre-disaster preparedness and post-disaster response stages, while explicitly accounting for the operational characteristics of drone-enabled deliveries and uncertainty in demand.

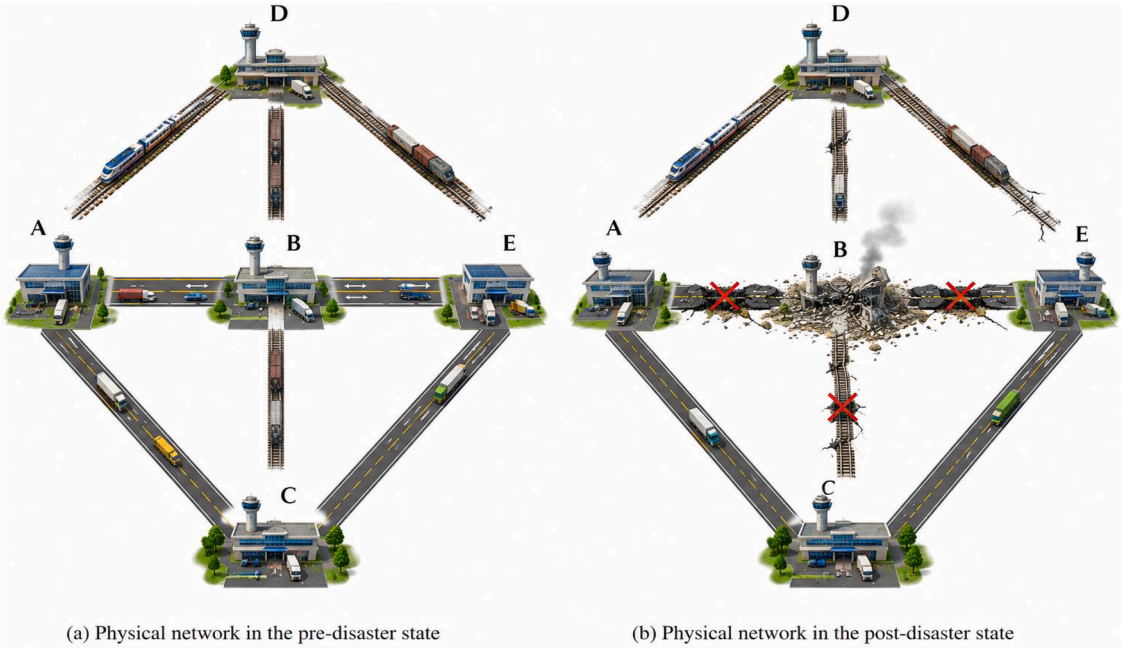
Overall, the disruption management literature has been largely developed within the context of humanitarian logistics, with a primary focus on facility protection, relief distribution, and the preservation of public services under extreme uncertainty. While this stream of research has significantly advanced the modeling of disrupted supply chains, it has largely overlooked the redesign of the SSND problem. In particular, existing studies rarely address how disruptions alter the underlying physical transportation network, defined by terminals and connecting arcs, and how scheduled services should be systematically reconfigured in response to the loss of terminals, arcs, and associated capacities. As a result, the interaction between physical network degradation and the redesign of scheduled services remains insufficiently explored, highlighting a critical gap between humanitarian disruption management models and SSND freight transportation planning.

2.3. Research gaps and contributions

Based on the above literature review on SSND and humanitarian logistics-oriented disruption management, several research gaps can be identified:

- Although uncertainty in demand and capacity has been widely studied in SSND models (Liu et al., 2023a), disruptions that explicitly damage terminals, arcs, and network connectivity have not been systematically incorporated into the SSND framework.
- Existing SSND studies predominantly assume a fixed underlying physical network and focus on service design and flow assignment. Consequently, the concept of post-disruption network redesign, where scheduled services must be reconfigured in response to the loss of terminals and arcs, remains largely unexplored.
- The disruption management literature in humanitarian logistics has made substantial contributions to facility protection, relief distribution, and recovery planning; however, these models are not designed to address freight-oriented SSND planning problems and do not capture the time-dependent demands and timetable of scheduled services.
- Most SSND formulations aim to minimize total system costs and implicitly assume full demand satisfaction. Only a limited number of studies consider demand selection decisions to maximize total profit (Taherkhani et al., 2022).
- While Benders decomposition (BD) has been successfully applied to classical SSND formulations to enhance computational efficiency (Mirzaei et al., 2025), its applicability and performance in solving SSND problems with explicit network redesign under disruption scenarios have not yet been investigated.

To address these gaps, this study makes the following contributions. First, it introduces the Scheduled Service Network Redesign (SSNR) problem as a novel extension of the classical SSND framework, explicitly capturing the impacts of disruptions on the physical transportation network. Second, the proposed SSNR model integrates demand selection decisions under disrupted conditions, allowing the freight transportation planners to prioritize profitable demand flows rather than assuming full demand fulfillment. Third, to overcome the computational complexity of the SSNR problem, a BD reformulation is developed that decomposes the proposed model into tractable subproblems. Finally, the effectiveness and computational efficiency of the proposed BD reformulation are systematically evaluated through numerical experiments and benchmarked against the exact solution solver.



(a) Physical network in the pre-disaster state (b) Physical network in the post-disaster state

Fig. 1. Graphical presentation of the physical network.

3. Problem definition

To comprehensively define the proposed SSNR problem, this article breaks it down into several essential components. First, the physical network under consideration, including its terminals and associated arcs, is described in Section 3.1. Second, the time-dependent demands, which form the core of the demand side, are outlined in Section 3.2. Third, the characteristics of the scheduled services, as the core of the supply side, are presented in Section 3.3. Finally, Section 3.4 elaborates on the planning levels associated with the SSNR problem. This section provides a detailed discussion of the assumptions and constraints related to time elements, which are critical for accurately modeling the SSNR problem.

3.1. Physical network

Fig. 1 illustrates the conceptual transportation network, comprising terminals and connecting arcs that facilitate the flow of demand. Formally, the physical network is defined as $\mathcal{G}^{\mathcal{PH}} = (\mathcal{N}^{\mathcal{PH}}, \mathcal{A}^{\mathcal{PH}})$, where $\mathcal{N}^{\mathcal{PH}}$ is the set of terminals, each with capacity u_n, $n \in \mathcal{N}^{\mathcal{PH}}$, and $\mathcal{A}^{\mathcal{PH}}$ is the set of arcs $a = (n, n')$ connecting distinct terminals $n \neq n'$. Terminals store and transfer demands to services across the network.

The physical network enables the organized and efficient movement of demands between terminals (Taherkhani et al., 2022). For instance, consider a pre-disaster network comprising five terminals, labeled A, B, C, D, and E, interconnected by multiple arcs, as shown in Fig. 1(a). This structure supports seamless demand transfers, ensuring that all demand origins and destinations align with existing terminals for cohesive and controlled flows. Terminals in a demand itinerary are categorized as origin, transfer, or destination terminals.

When a disaster occurs, specific terminals and arcs within the network may be disrupted. For example, a disaster that destroys terminal B results in the deactivation of all arcs directly associated with this terminal, effectively isolating it from the network. Fig. 1(b) illustrates the post-disaster state, where all connections to terminal B are deactivated, disrupting service paths that previously included this terminal. Consequently, demands with terminal B as an origin or destination remain unsatisfied, underscoring the vulnerability of the network to disruptions and the significant impact on demand fulfillment.

To model the post-disaster impacts, z_n is introduced as a binary parameter that indicates whether a terminal $n \in \mathcal{N}^{\mathcal{PH}}$ has been destroyed, $z_n = 1$, or remains operational, $z_n = 0$. Similarly, if only specific arcs are disrupted, s_a is used as a binary parameter for each arc $a \in \mathcal{A}^{\mathcal{PH}}$, where $s_a = 1$ denotes a disrupted arc and $s_a = 0$ denotes an undisturbed arc.

The process of demand movement within this network follows a structured demand itinerary. Shipments are initially picked up from customers at an origin terminal, where incoming loads are either transferred through unloading, cross-docking, and reloading onto outgoing vehicles or classified for sorting and reloading at transfer terminals (Crainic and Rei, 2025). These shipments are subsequently dispatched to their designated destination terminals. Upon arrival, freight is segregated and reconsolidated for local delivery, ensuring efficient end-to-end service. Finally, the operational tasks associated with local pickup and delivery at origin and destination terminals fall under operational planning and are beyond the scope of the SSNR problem in this study.

3.2. Time-dependent demands

In the SSNR problem for freight transportation, each demand $k \in \mathcal{K}$ is associated with a predefined origin $o_k \in \mathcal{N}^{\mathcal{PH}}$ and destination $d_k \in \mathcal{N}^{\mathcal{PH}}$, where $o_k \neq d_k$. A fundamental assumption of the SSNR framework is that all demands are specified based on the pre-disaster state of the physical network. Consequently, the demand definition and planning processes are conducted without prior knowledge of potential disruptions affecting the network infrastructure.

These demands represent sets of shipments with a total volume q_k that require uniform packing, loading, transportation, and warehousing processes. Such uniformity ensures compatibility for joint loading, simplifying the handling of shipments in freight transportation. Additionally, the divisibility of q_k allows demands to be split across multiple scheduled services if necessary, providing greater flexibility in the network. When a specific demand $k \in \mathcal{K}$ is selected by the proposed SSNR model, the corresponding revenue per unit volume, denoted as θ_k, is incorporated into the SSNR model to maximize the total profit generated from fulfilling demands.

Handling costs are another critical component of the demand-side considerations in the SSNR problem. For each demand k, the unit handling and holding cost at a terminal $n \in \mathcal{N}^{\mathcal{PH}}$ is represented by c_{kn}. This cost encompasses expenses related to warehousing as well as potential delays incurred at the terminals. These costs must be carefully managed to maintain the economic efficiency of the network while meeting the service requirements of the selected demands.

Time-dependent attributes of demands are essential for ensuring timely and efficient service. Each demand k is associated with a release time α_k, which specifies when the demand becomes available for pickup at its origin terminal. Furthermore, customers express a preferred delivery time β_k, representing the ideal time for the shipment's arrival at the destination terminal. To account for scheduling flexibility, a time window is defined for each demand, bounded by an earliest due date β_k^- and a latest due date β_k^+. The former indicates the earliest acceptable delivery time, while the latter represents the latest permissible time for delivery to avoid penalties or service disruptions. To comply with the time window constraints, any selected demand $k \in \mathcal{K}$ should be delivered within the interval $[\beta_k^-, \beta_k^+]$, satisfying $\beta_k^- \leq \beta_k \leq \beta_k^+$.

3.3. Scheduled services

Scheduled services, such as trains or intermodal freight lines, represent the supply side of the SSNR problem and facilitate the transportation of demand between terminals (Crainic et al., 2018). A set of scheduled services, denoted as $\sigma \in \Sigma$, is used to define these services. Each scheduled service $\sigma \in \Sigma$ operates along a specific route, starting from an origin terminal o_σ and terminating at a destination terminal d_σ, where $o_\sigma, d_\sigma \in \mathcal{N}^{\mathcal{PH}}$ and $o_\sigma \neq d_\sigma$. These services can either follow a single physical arc or be composed of a sequence of connected arcs, as described by the path $\Pi(\sigma)$, which consists of legs $\pi_i(\sigma)$. Each leg represents a direct connection between two consecutive terminals, such that $\pi_i(\sigma) \in \mathcal{A}^{\mathcal{PH}}$ for each $i = 1, 2, \ldots, |\Pi(\sigma)|$, where $|\Pi(\sigma)|$ denotes the number of legs in the scheduled service σ. This sequence of legs forms a stepwise transport route across the network, enabling the transport of selected demands.

The main assumption is that these scheduled services are defined based on a disruption-free physical network. At the time of defining and planning the services, the service planners are assumed to have no prior knowledge of potential disruptions affecting the network.

Each scheduled service is associated with a defined capacity, u_σ, which may vary across different legs of the service. Specifically, the capacity for each leg $\pi_i(\sigma)$ is denoted as $u(\pi_i(\sigma))$. This variation allows for flexibility in accommodating the demands of varying volumes across different parts of the route. The arcs in the physical network, $a \in \mathcal{A}^{\mathcal{PH}}$, are linked to a scheduled service $\sigma(a) \in \Sigma$, indicating which scheduled service is responsible for the transportation along with the arc a. The terminals connected by each arc $a \in \mathcal{A}^{\mathcal{PH}}$ are denoted as $n(a)$, representing the origin or destination terminal within the arc.

Each scheduled service follows a predefined timetable that specifies the departure and arrival times for each leg. For each leg $\pi_i(\sigma)$, the departure time from the origin terminal is denoted by $\alpha(\pi_i(\sigma))$, and the arrival time at the destination terminal is denoted by $\beta(\pi_i(\sigma))$. The duration of each leg $\pi_i(\sigma)$ is computed as the absolute difference between the arrival and departure times, expressed as $\tau(\pi_i(\sigma)) = |\beta(\pi_i(\sigma)) - \alpha(\pi_i(\sigma))|$. This transit time represents the time required to move between the origin and destination terminals for each leg of the scheduled service.

From an economic perspective, each scheduled service has a pricing structure that includes two key cost components. The first is a fixed cost, f_σ, which is incurred when the service is set up. This fixed cost is independent of the volume of demand transported. The second component is a unit transportation cost, denoted by $c_{k\sigma}$, which applies to each unit volume of demand transported over each leg of the service.

3.4. Planning levels

The proposed SSNR problem operates across two distinct planning levels: strategic and tactical. At the strategic level, the focus is on long-term decisions that define the core structure of the network (Seeve and Vilkkumaa, 2022). In the context of the SSNR, the physical network in a post-disaster state represents the elements that are disrupted, such as terminals and arcs, which cause high-impact, long-term effects on demand itineraries and scheduled services. Although decisions of the SSNR are not made directly on the physical network structure, the transition from the pre-disaster to the post-disaster network introduces a set of parameters that significantly influence strategic planning. In contrast, tactical planning addresses medium-term objectives, typically spanning a year or a specific season (Megahed and Goetschalckx, 2018). At this level, the decisions revolve around selecting which demands to fulfill and assigning the optimal services to meet these demands, even within a disrupted network.

The planning horizon for the SSNR problem is aligned with the chosen planning level, whether strategic or tactical. The horizon is further divided into equal intervals, referred to as the scheduled length, denoted by $\mathcal{T}$. These intervals may represent weekly, monthly, or other suitable time periods depending on the specific needs of the network. Within each scheduled length, it is assumed that demands follow a predictable, cyclical pattern, enabling the formulation of more accurate and practical scheduling decisions. Additionally, the scheduled length $\mathcal{T}$ is partitioned into finer time periods, denoted as $t \in \mathcal{T} = \{1, 2, \dots, T\}$, which are typically defined on an hourly or daily basis. This finer granularity allows for precise planning of time-dependent parameters and variables in the SSNR problem, such as service arrival and departure times, as well as demand processing.

A key assumption within the scheduled length $\mathcal{T}$ is the allowance for the use of backward arcs in the scheduled services. This assumption addresses cases where a service's arrival time at a terminal extends beyond the current scheduled length. By permitting certain arcs to reverse their direction, the SSNR problem ensures that service departures remain within the defined scheduled length $\mathcal{T}$, maintaining the continuity of the service scheduling framework.

Another important consideration within the scheduled length is the absence of transfer costs for simultaneous service transitions. Specifically, if a demand arrives at a terminal from one service at time $t \in \mathcal{T}$, it can immediately transition to another service departing at the same time t without incurring any additional transfer costs. However, if the next service departs at a later time $t' \in \mathcal{T}$, where $t' > t$, the demand incurs a holding cost for the duration of its stay at the terminal.

To manage timely deliveries and mitigate the impact of delays, penalty costs are incorporated for meeting delivery time windows within each scheduled length. While delays are permitted, they incur penalty costs, denoted as ρ_k for each demand $k \in \mathcal{K}$, to uphold contractual requirements and meet customer expectations. These penalties incentivize the timely fulfillment of demand while acknowledging the possibility of delays. Specifically, when a demand k has a delivery window defined by $\beta_k \le \beta_\sigma \le \beta_k^+$, where β_σ is the actual delivery time, the adjusted unit transportation cost for service σ becomes $\bar{c}_{k\sigma} = c_{k\sigma} + \rho_k$. In addition, if $\beta_\sigma < \beta_k^-$, the corresponding solution is labeled by an infinite cost to the objective function.

4. Mathematical models

In this section, we first introduce the time-space network representation for the SSNR problem, which serves as the foundational modeling framework (Section 4.1). Subsequently, we present the classical SSND model and extend it to develop the SSNR formulation for the redesign of scheduled services, explicitly accounting for disaster-induced disruptions (Section 4.2).

4.1. Time-space network

Considering the time-dependent nature of parameters and variables in freight transportation, a time-space network is employed as the modeling framework (Taherkhani et al., 2022). This time-space network enables the formulation of the SSNR problem to establish both strategic and tactical plans, where events and decisions are explicitly captured across discrete time periods.

Following established conventions in the literature (Crainic et al., 2021), we extend the physical network $\mathcal{G}^{\mathcal{PH}} = (\mathcal{N}^{\mathcal{PH}}, \mathcal{A}^{\mathcal{PH}})$ into a time-space graph, denoted as $\mathcal{G} = (\mathcal{N}, \mathcal{A})$. The schedule length is discretized into a set of time periods, $\mathcal{T} = \{1, 2, \dots, T\}$. In this framework, a terminal is characterized by its physical location $n \in \mathcal{N}^{\mathcal{PH}}$ and the associated time period $t \in \mathcal{T}$. Each terminal in the time-space network is represented as $(n, t) \in \mathcal{N}$, signifying its spatial and temporal coordinates.

The arcs in the time-space network are divided into two categories: traveling arcs and holding arcs, collectively defined as $\mathcal{A} = \mathcal{A}^{\mathcal{MO}} \cup \mathcal{A}^{\mathcal{HO}}$. A traveling arc corresponds to a service leg in the physical network, denoted by $\pi_i(\sigma(a)) = (n, n') \in \mathcal{A}^{\mathcal{PH}}$, where $n, n' \in \mathcal{N}^{\mathcal{PH}}$ and $n \neq n'$. Each service leg has a duration $\tau(\pi_i(\sigma(a)))$ and connects time periods t and t' in the time-space network, where $t < t'$. This service leg is represented as $\sigma(a) = \alpha = ((n, t), (n', t')) \in \mathcal{A}^{\mathcal{MO}}$ within the time-space graph.

In addition to traveling arcs, the time-space network incorporates holding arcs, denoted as $\mathcal{A}^{\mathcal{HO}}$. A holding arc represents the inventory status or waiting at a terminal. For a terminal $n \in \mathcal{N}^{\mathcal{PH}}$, a holding arc connects consecutive time periods, expressed as $a = ((n, t), (n, t+1)) \in \mathcal{A}^{\mathcal{HO}}$. The terminal associated with a holding arc is indicated by $n(a)$ for any $a \in \mathcal{A}^{\mathcal{HO}}$.

Using these constructs, we define two key sets of arcs for each terminal $(n, t) \in \mathcal{N}$. The set $\mathcal{A}^{(+)}(n, t)$ includes all traveling and holding arcs that originate from the terminal at time t, while $\mathcal{A}^{(-)}(n, t)$ includes all arcs converging at the terminal at time t. These definitions provide the basis for flow conservation and decision modeling in the time-space network representation of the SSNR problem.

To numerically illustrate the time-space network, consider an example where a demand's itinerary begins at its origin terminal, denoted as $o_k \in \mathcal{N}^{\mathcal{PH}}$. At any given time $t \in \mathcal{T}$, the demand has two options: it may either remain at its origin terminal using a holding arc, represented as $a = ((o_k, t), (o_k, t+1)) \in \mathcal{A}^{\mathcal{HO}}$, or move to a transfer terminal $n \in \mathcal{N}^{\mathcal{PH}}$ via a scheduled service, $\sigma(a) \in \Sigma$. Upon reaching a transfer terminal, the demand can again choose to either remain at the terminal via a holding arc, $a = ((n, t''), (n, t''+1)) \in \mathcal{A}^{\mathcal{HO}}$, or proceed to another transfer terminal or directly to its destination terminal, denoted as $d_k \in \mathcal{N}^{\mathcal{PH}}$.

For example, consider the pre-disaster and post-disaster states of the physical network depicted in Fig. 1. The corresponding time-space network, which illustrates three complete itineraries, is presented in Fig. 2. In this numerical example, the schedule length spans 8 time periods, denoted as $|\mathcal{T}| = 8$. The network includes 5 terminals: A, B, C, D, and E. To further contextualize this example, data for three demands is provided in Table 1. This information encompasses the origins, destinations, and other time attributes of the demands, enabling a detailed analysis of the time-space network's representation and functionality.

To clarify the numerical example for the time-space network illustrated in Fig. 2, it is important to highlight that there is no demand itinerary for the second demand as it was not selected. The destination of this demand was terminal B, which is unavailable

Table 1
Demands in the numerical example for the time-space network.

Demand	o_k	d_k	α_k	β_k	β_k^-	β_k^+
$k = 1$	Terminal A	Terminal D	$t = 1$	$t = 7$	$t = 6$	$t = 8$
$k = 2$	Terminal C	Terminal B	$t = 2$	$t = 7$	$t = 6$	$t = 8$
$k = 3$	Terminal E	Terminal A	$t = 2$	$t = 7$	$t = 6$	$t = 8$

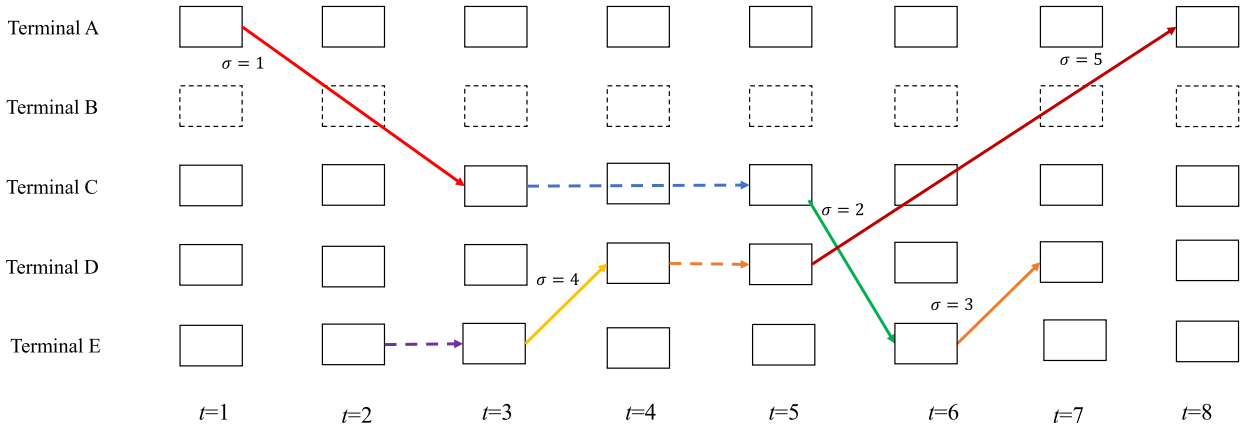


Fig. 2. An example of demand itineraries in the time-space network.

in the post-disaster state. In this regard, the optimization model selected only the first and third demands. The itineraries for these selected demands from Table 1 are as follows:

For the first selected demand ($k = 1$), the demand becomes available at time $t = 1$. Although there are direct connections between terminal A and terminal D, which serve as the origin and destination terminals respectively for this demand, there is no scheduled service available for transportation between these terminals at this time till time $t = 5$. As a result, the optimization model selects the scheduled service $\sigma = 1$ to transport the first demand from terminal A to terminal C. This demand arrives at terminal C at time $t = 3$. Since no service is available for transporting this demand immediately upon arrival, it is held at terminal C from time $t = 3$ to $t = 5$. Subsequently, the demand is transported via scheduled service $\sigma = 2$ from terminal C to terminal E, with the transit occurring from $t = 5$ to $t = 6$. Upon arrival at terminal E, the demand is immediately transported using scheduled service $\sigma = 3$ from terminal E to terminal D, completing its journey at $t = 7$. As the demand arrives on time at its destination, i.e., terminal D, at $t = 7$, no penalty cost is incurred for this selected demand.

Lastly, for the third demand ($k = 3$), the demand becomes available at time $t = 2$ at terminal E, which serves as the origin. Since there is no scheduled service available at this time, the demand is held at terminal E from time $t = 2$ to $t = 3$. It is then transported via the scheduled service $\sigma = 4$, transferring it from terminal E to terminal D at time $t = 4$. Upon arrival, the demand is held at terminal D from time $t = 4$ to $t = 5$. Subsequently, it is transported from terminal D at time $t = 5$ to terminal A at time $t = 8$ using the scheduled service $\sigma = 5$. Since the preferred delivery time for this demand is $t = 7$, a penalty cost is incurred and added to the unit transportation cost of the last scheduled service, resulting in $c_{3,5} + \rho_3$.

4.2. Mixed-integer programs

To highlight the key distinctions between the SSND and SSNR formulations, we first present the classical SSND model based on the recent review by Crainic and Rei (2025) in Section 4.2.1. Subsequently, the proposed MIP formulation for the SSNR problem is developed in Section 4.2.2.

4.2.1. Classical SSND

To define the classical SSND (Crainic and Rei, 2025), the following binary and continuous decision variables are defined:

$$y_\sigma = \begin{cases} 1, & \text{if scheduled service } \sigma \in \Sigma \text{ is selected,} \\ 0, & \text{otherwise;} \end{cases}$$

x_{ak} Amount of demand $k \in \mathcal{K}$ on arc $a \in \{\mathcal{A}^{\mathcal{MO}} \cup \mathcal{A}^{\mathcal{HO}}\}$.

Using the above variables, the MIP model for classical SSND is defined as follows:

$$\min\Big(\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}^{\mathcal{PH}}} c_{kn(a)}x_{ak} + \sum_{\sigma\in\Sigma} f_\sigma y_\sigma + \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{MO}}} \bar{c}_{ka}x_{ak}\Big) \tag{1}$$

s.t.

$$\sum_{a\in\mathcal{A}^{(+)}(o_k,\alpha_k)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(o_k,\alpha_k)} x_{ak} = q_k \qquad \forall k \in \mathcal{K} \tag{2}$$

$$\sum_{a\in\mathcal{A}^{(+)}(d_k,\beta_k^+)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(d_k,\beta_k^+)} x_{ak} = -q_k \qquad \forall k \in \mathcal{K} \tag{3}$$

$$\sum_{a\in\mathcal{A}^{(+)}(n,t)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(n,t)} x_{ak} = 0 \qquad \forall k \in \mathcal{K}, t \in \mathcal{T}, n \in \mathcal{N}^{\mathcal{PH}} \tag{4}$$

$$\sum_{k\in\mathcal{K}} x_{ak} \le u_{\sigma(a)} y_\sigma \qquad \forall \sigma \in \Sigma, a \in \mathcal{A}^{\mathcal{MO}} \tag{5}$$

$$\sum_{k\in\mathcal{K}} x_{ak} \le u_{n(a)} \qquad \forall n \in \mathcal{N}^{\mathcal{PH}}, a \in \mathcal{A}^{\mathcal{HO}} \tag{6}$$

$$y_\sigma \in \{0,1\} \qquad \forall \sigma \in \Sigma \tag{7}$$

$$x_{ak} \ge 0 \qquad \forall k \in \mathcal{K}, a \in \mathcal{A}^{\mathcal{MO}} \cup \mathcal{A}^{\mathcal{HO}} \tag{8}$$

The objective function for the SSND problem, presented in Eq. (1), aims to minimize the total cost by considering the holding costs at terminals, fixed costs for selecting scheduled services, and transportation costs for moving demands along selected services.

Constraint set (2) enforces flow balance at the origin for each demand, ensuring that the outgoing flow matches the demand's availability at its origin terminal. Similarly, constraint set (3) ensures flow balance at the destination terminal for each demand, guaranteeing that all demand flows are delivered completely. Constraint set (4) maintains overall flow balance at transfer terminals, ensuring that the total inflow matches the total outflow across the network's intermediate nodes.

Constraint set (5) enforces capacity limitations on traveling arcs, ensuring that the total demand flow on any traveling arc does not exceed its capacity when associated services are selected. Constraint set (6) ensures that the total demand handled at the terminals does not exceed their holding capacities. Relations (7) and (8) define the feasible range for decision variables, specifying binary variables for service selection and continuous variables for demand flow quantities.

4.2.2. Proposed SSNR

In the SSNR formulation, in addition to the decision variables x_{ak} and y_σ similar to the classical SSND, the following binary variable is defined:

$$w_k = \begin{cases} 1, & \text{if demand } k \in \mathcal{K} \text{ is selected to be satisfied,} \\ 0, & \text{otherwise;} \end{cases}$$

Using the above variables, the MIP model for the proposed SSNR is defined as follows:

$$\max\Big(\Big(\sum_{k\in\mathcal{K}} \theta_k q_k w_k\Big) - \Big(\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}^{\mathcal{PH}}} c_{kn(a)} x_{ak} + \sum_{\sigma\in\Sigma} f_\sigma y_\sigma + \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{MO}}} \bar{c}_{ka} x_{ak}\Big)\Big) \tag{9}$$

s.t.

$$y_\sigma + z_n \le 1 \qquad \forall n \in \mathcal{N}^{\mathcal{PH}}, (n,n') \in \Pi(\sigma), n \ne n', \sigma \in \Sigma \tag{10}$$

$$y_\sigma + s_a \le 1 \qquad \forall a \in \mathcal{A}^{\mathcal{MO}}, a \in \Pi(\sigma), \sigma \in \Sigma \tag{11}$$

$$x_{ak} \le q_k(1 - z_{n(a)}) \qquad \forall k \in \mathcal{K}, n \in \mathcal{N}^{\mathcal{PH}}, a \in \mathcal{A}^{\mathcal{HO}} \tag{12}$$

$$\sum_{a\in\mathcal{A}^{(+)}(o_k,\alpha_k)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(o_k,\alpha_k)} x_{ak} = w_k q_k \qquad \forall k \in \mathcal{K} \tag{13}$$

$$\sum_{a\in\mathcal{A}^{(+)}(d_k,\beta_k^+)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(d_k,\beta_k^+)} x_{ak} = -w_k q_k \qquad \forall k \in \mathcal{K} \tag{14}$$

$$\sum_{a\in\mathcal{A}^{(+)}(n,t)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(n,t)} x_{ak} = 0 \qquad \forall k \in \mathcal{K}, t \in \mathcal{T}, n \in \mathcal{N}^{\mathcal{PH}} \tag{15}$$

$$\sum_{k\in\mathcal{K}} x_{ak} \le u_{\sigma(a)} y_\sigma \qquad \forall \sigma \in \Sigma, a \in \mathcal{A}^{\mathcal{MO}} \tag{16}$$

$$\sum_{k\in\mathcal{K}} x_{ak} \le (1 - z_{n(a)}) u_{n(a)} \qquad \forall n \in \mathcal{N}^{\mathcal{PH}}, a \in \mathcal{A}^{\mathcal{HO}} \tag{17}$$

$$w_k \in \{0,1\} \qquad \forall k \in \mathcal{K} \tag{18}$$

$$y_\sigma \in \{0,1\} \qquad \forall \sigma \in \Sigma \tag{19}$$

$$x_{ak} \ge 0 \qquad \forall k \in \mathcal{K}, a \in \mathcal{A}^{\mathcal{MO}} \cup \mathcal{A}^{\mathcal{HO}} \tag{20}$$

The objective function for the SSNR problem, presented in Eq. (9), aims to maximize profit by considering the total revenue minus the total cost. It includes four components: the revenue generated from selecting demands, holding costs at terminals, fixed costs for

activating scheduled services, and transportation costs for moving demands along selected services. By maximizing the difference between revenue and costs, the model optimizes the network efficiency in a post-disaster state.

Constraint set (10) ensures the operational feasibility of scheduled services by linking their activation to the availability of associated terminals, which must remain undisrupted by a disaster. Constraint set (11) connects service legs to physical arcs, ensuring that a scheduled service can only operate if all corresponding physical arcs are functional and undisrupted. Constraint set (12) relates the activation of terminals to demand flows on holding arcs, ensuring that demands can only be stored at terminals that are available and undamaged.

Constraint set (13) enforces flow balance at the origin for each selected demand, ensuring that the outgoing flow matches the demand's availability at its origin terminal. Similarly, constraint set (14) ensures flow balance at the destination for each selected demand, guaranteeing that all demand flows are delivered completely. Constraint set (15) maintains overall flow balance at transfer terminals, ensuring that the total inflow matches the total outflow across the network's intermediate nodes.

Constraint set (16) enforces capacity limitations on traveling arcs, ensuring that the total demand flow on any arc does not exceed its capacity when associated services are selected. Constraint set (17) ensures that the total demand handled at active terminals does not exceed their holding capacities, linking storage capacity to terminal activation. Relations (18) to (20) define the feasible range for decision variables, specifying binary variables for activation decisions and continuous variables for flow quantities, thus completing the MIP model's definition.

Based on a comparison between the classical SSND model given in Section 4.2.1 and the proposed SSNR model, the primary contributions of the SSNR model can be summarized as follows:

- Unlike SSND, which typically focuses on cost minimization for the fixed and variable costs of the services and terminals as given in Eq. (1), the SSNR model introduces total profit maximization as given in Eq. (9) and demand selection as novel elements, explicitly incorporating revenue-oriented considerations within the SSND framework.
- Constraints (10) to (12) in the SSNR model are specifically designed to capture disruptions affecting terminals and services. These aspects are entirely original and have not been addressed in classical SSND studies.
- The SSNR model redefines the network flow conservation constraints typically used in SSND. Specifically, by comparing constraints (2) and (3) in the classical SSND with constraints (13) and (14) in the SSNR, the latter explicitly integrates demand selection while determining demand itineraries. This allows the SSNR model to prioritize or defer demands based on post-disruption conditions.
- The SSNR model introduces a novel formulation for terminal capacity that explicitly accounts for disruptions. Specifically, constraint set (17) in the SSNR model, compared to constraint set (6) in the classical SSND, enforces capacity limitations in response to terminal disruptions, i.e., an aspect not addressed in previous SSND formulations.

Overall, the SSNR model extends classical SSND by incorporating post-disruption decision-making, profit-oriented objectives, demand prioritization, and explicit disruption-aware capacity constraints, providing a more comprehensive framework for resilient freight transportation network planning.

5. BD Reformulation algorithm

Given the structural complexity of the proposed SSNR formulation, this study adopts a standard BD approach as a solution method, following well-established practices in fixed-charge network design problems (Costa, 2005; Rahmaniani et al., 2017). The BD method proposed by Benders (1962) is a widely used decomposition approach for solving MIP problems where both integer and continuous variables are present (Fathollahi-Fard et al., 2023). The main idea is to decompose the original problem into two interconnected models: a Master Problem (MP) and a Primary Subproblem (PS). The choice of BD is motivated by the intrinsic separation between strategic binary decisions and operational continuous decisions embedded in the SSNR model. Specifically, the SSNR formulation naturally decomposes into:

- a set of binary variables governing service activation, and demand acceptance decisions, and
- a set of continuous flow variables that determine feasible demand itineraries over the disrupted time-space network.

Fixing the binary decisions results in a linear programming subproblem that evaluates network feasibility and computes the itinerary costs of accepted demands. This structure aligns directly with the classical BD formulation, where the MP captures the combinatorial design decisions, and the PS evaluates network feasibility and cost implications through a linear flow model. The overall structure of the BD reformulation for the SSNR is illustrated in Fig. 3.

The overall solution process is organized as follows. First, a linear relaxation is introduced to generate an initial solution, which provides the foundation for constructing the PS model (Section 5.1). Next, the PS model and its dual formulation are developed to derive feasibility and optimality cuts (Section 5.2). The MP formulation is then presented (Section 5.3). Finally, the iterative BD algorithm is described in detail (Section 5.4).

5.1. Linear relaxation

To generate the initial solution, this study adopts a linear relaxation approach, a widely used method in the literature on network design problems (Crainic and Gendreau, 2007; Fathollahi-Fard and Li, 2026). In this approach, the binary decision variables y_σ (service selection) and w_k (demand selection) are relaxed to continuous variables. This transformation converts the original MIP

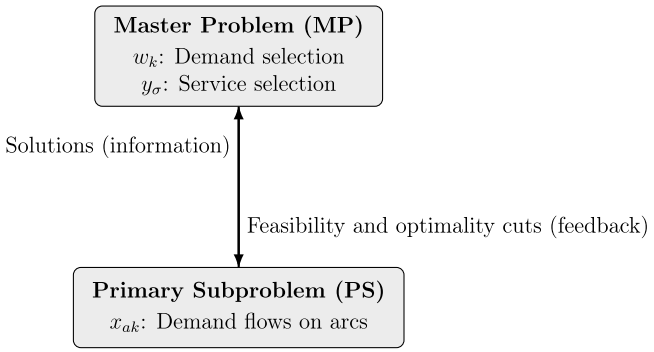


Fig. 3. BD reformulation for the SSNR problem.

model into a linear programming (LP) formulation, providing a linear relaxation of the proposed SSNR model. The relaxed model is formulated as follows:

$$\max\left(\sum_{k\in\mathcal{K}}\theta_k q_k w_k - \left(\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}^{\mathcal{PH}}} c_{kn(a)}x_{ak} + \sum_{\sigma\in\Sigma} f_\sigma y_\sigma + \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{MO}}}\bar{c}_{ka}x_{ak}\right)\right) \tag{21}$$

s.t.

Constraints (10) to (17) and (20),

$$0 \le w_k \le 1 \qquad \forall k \in \mathcal{K}, \tag{22}$$

$$0 \le y_\sigma \le 1 \qquad \forall \sigma \in \Sigma. \tag{23}$$

After solving the relaxed LP model using an exact solver such as CPLEX, any variable y_σ or w_k with a non-zero value is initially rounded up to one, yielding a candidate integer solution. Since this rounding step does not, by itself, guarantee feasibility with respect to capacity constraints of the services (constraint set (17)), a feasibility-repair heuristic is subsequently applied (Fathollahi-Fard and Li, 2026). Specifically, the heuristic adjusts the selected services and/or accepted demands to eliminate capacity violations on scheduled services, ensuring that all constraints of the original SSNR model are satisfied. The total profit associated with the repaired solution is then evaluated, resulting in an initial feasible solution denoted by Sol_0, which is also set as the current best-known solution, Sol_{best}. This feasible solution provides the initial values of y_σ and w_k for solving the PS model.

5.2. Primary subproblem

Based on the current best-known solution, Sol_{best}, the BD reformulation defines the fixed values $\bar{y}_\sigma = y_\sigma$ and $\bar{w}_k = w_k$. In this regard, the PS model, which focuses on optimizing the demand flow, is formulated as follows:

$$\min\left(\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}^{\mathcal{PH}}} c_{kn(a)}x_{ak} + \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{MO}}}\bar{c}_{ka}x_{ak}\right) \tag{24}$$

s.t.

Constraints (12), (15), (17) and (20)

$$\sum_{a\in\mathcal{A}^{(+)}(o_k,\alpha_k)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(o_k,\alpha_k)} x_{ak} = \bar{w}_k q_k \qquad \forall k \in \mathcal{K} \tag{25}$$

$$\sum_{a\in\mathcal{A}^{(+)}(d_k,\beta_k^+)} x_{ak} - \sum_{a\in\mathcal{A}^{(-)}(d_k,\beta_k^+)} x_{ak} = -\bar{w}_k q_k \qquad \forall k \in \mathcal{K} \tag{26}$$

$$\sum_{k\in\mathcal{K}} x_{ak} \le u_{\sigma(a)}\bar{y}_\sigma \qquad \forall \sigma \in \Sigma, a \in \mathcal{A}^{\mathcal{MO}} \tag{27}$$

If the given fixed values $\bar{y}_\sigma = y_\sigma$ and $\bar{w}_k = w_k$ are feasible based on the SSNR model, the PS model is also feasible. Then, the BD reformulation needs to compute the dual of the PS model (DS). In this regard, the DS model defines the following variables:

- $\varphi_k(n,t)$: dual variable associated with the flow-balance equality for constraints (15), (25) and (26) representing the terminal n and time t for demand k.
- β_a: dual variable for constraint set (27).
- γ_a: dual variable for constraint set (17).
- δ_{ak}: dual variable for constraint set (12).

From the dual perspective of the SSNR model, for each arc $a \in \mathcal{A}$, the DS model defines $(n_a^{\text{tail}}, t_a^{\text{tail}})$ as the node-time pair corresponding to the start of arc a, and $(n_a^{\text{head}}, t_a^{\text{head}})$ as the node-time pair corresponding to the end of arc a. This notation allows a precise representation of the temporal and spatial positions associated with each arc within the network, which is essential for formulating the DS constraints and understanding the demand flow across the network.

Based on the DS model, a lower bound is estimated as the optimality cut for the BD reformulation:

$$M \geq \sum_{k\in\mathcal{K}} \bar{w}_k q_k \Big(\varphi_k(o_k,\alpha_k) - \varphi_k(d_k,\beta_k^+)\Big) - \sum_{a\in\mathcal{A}^{\mathcal{MO}}} u_{\sigma(a)}\bar{y}_\sigma \beta_a - \sum_{a\in\mathcal{A}^{\mathcal{HO}}} (1-z_{n(a)})u_{n(a)}\gamma_a - \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{HO}}} q_k(1-z_{n(a)})\delta_{ak} \tag{28}$$

To ensure that the PS model is feasible, the following feasibility cut must be satisfied:

$$0 \geq \sum_{k\in\mathcal{K}} \bar{w}_k q_k \Big(\varphi_k(o_k,\alpha_k) - \varphi_k(d_k,\beta_k^+)\Big) - \sum_{a\in\mathcal{A}^{\mathcal{MO}}} u_{\sigma(a)}\bar{y}_\sigma \beta_a - \sum_{a\in\mathcal{A}^{\mathcal{HO}}} (1-z_{n(a)})u_{n(a)}\gamma_a - \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{HO}}} q_k(1-z_{n(a)})\delta_{ak} \tag{29}$$

5.3. Master problem

After solving the DS model, the optimal values of the dual variables $\bar{\varphi}_k(n,t)$, $\bar{\beta}_a$, $\bar{\gamma}_a$, and $\bar{\delta}_{ak}$ are passed to the MP model to construct the corresponding optimality and feasibility cuts, which are then used to generate a new solution within the BD framework. In this regard, the MP is formulated using the optimality and feasibility cuts derived from the SP as follows:

$$\max\left(\sum_{k\in\mathcal{K}} \theta_k q_k w_k - \Big(\sum_{\sigma\in\Sigma} f_\sigma y_\sigma + M\Big)\right) \tag{30}$$

s.t.

Constraints (10), (11), (18), and (19),

$$M \geq \sum_{k\in\mathcal{K}} w_k q_k \Big(\bar{\varphi}_k(o_k,\alpha_k) - \bar{\varphi}_k(d_k,\beta_k)\Big) - \sum_{a\in\mathcal{A}^{\mathcal{MO}}} u_{\sigma(a)} y_\sigma \bar{\beta}_a - \sum_{a\in\mathcal{A}^{\mathcal{HO}}} (1-z_{n(a)})u_{n(a)}\bar{\gamma}_a - \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{HO}}} q_k(1-z_{n(a)})\bar{\delta}_{ak} \tag{31}$$

$$0 \geq \sum_{k\in\mathcal{K}} w_k q_k \Big(\tilde{\varphi}_k(o_k,\alpha_k) - \tilde{\varphi}_k(d_k,\beta_k^+)\Big) - \sum_{a\in\mathcal{A}^{\mathcal{MO}}} u_{\sigma(a)} y_\sigma \tilde{\beta}_a - \sum_{a\in\mathcal{A}^{\mathcal{HO}}} (1-z_{n(a)})u_{n(a)}\tilde{\gamma}_a - \sum_{k\in\mathcal{K}}\sum_{a\in\mathcal{A}^{\mathcal{HO}}} q_k(1-z_{n(a)})\tilde{\delta}_{ak} \tag{32}$$

Although the above MP model remains an integer program compared to the original MIP of the SSNR model, it contains fewer constraints and eliminates complex ones, such as capacity constraints. This reduction enhances the tractability of the SSNR problem, making the BD reformulation more efficient for solving complex network design problems such as SSNR.

5.4. Iterative algorithm

The BD reformulation is an iterative procedure designed to efficiently solve the SSNR problem by decomposing it into two interconnected components: the MP and the PS. Then, the DS model is extracted from the PS model. The procedure begins with an initial feasible solution obtained from the linear relaxation of the SSNR. At each iteration, the DS is solved using the current service design and demand selection decisions. From the DS, feasibility and optimality cuts are generated and added to the MP to refine its representation of the original SSNR problem. The MP is then re-optimized to update the solution. If no improvement is achieved or the solution stabilizes, the algorithm terminates. Otherwise, the updated solution is used to continue the iterative process. The iterative process of the BD reformulation is shown in Algorithm 1.

6. Computational experiments

This section begins by generating the test instances used to evaluate the proposed BD reformulation algorithm and the SSNR model in Section 6.1. These instances form the foundation for assessing the model's performance and ensuring its practical applicability in real-world disaster scenarios. Section 6.2 examines the computational efficiency of the BD reformulation by comparing its performance against the exact solver. Subsequently, Section 6.3 introduces the performance metrics, which are essential for evaluating the efficiency and effectiveness of the SSNR model in both pre-disaster and post-disaster settings. In Section 6.4, a detailed analysis of the proposed SSNR model is conducted using various test instances to compare outcomes under pre-disaster and post-disaster conditions. Finally,

Algorithm 1 BD Reformulation algorithm for SSNR.

Initialize $Sol_{\text{best}} \leftarrow Sol_0$ (from SSNR linear relaxation)
Set iteration counter $it \leftarrow 0$
Set maximum iterations $MaxIt$
Set termination flag $r \leftarrow 0$
while $r = 0$ **do**
    Solve DS using service design y_σ and demand selection w_k from Sol_{best}
    Generate feasibility and optimality cuts from DS and add them to MP
    Solve MP to obtain objective value Z
    if $Z = Sol_{\text{best}}$ **then**
        Set $r \leftarrow 1$ ▷ Convergence achieved
    else if $Z > Sol_{\text{best}}$ **then**
        Update $Sol_{\text{best}} \leftarrow Z$
    end if
    Increment iteration counter: $it \leftarrow it + 1$
    if $it = MaxIt$ **then**
        Set $r \leftarrow 1$ ▷ Maximum iterations reached
    end if
end while
return Sol_{best}

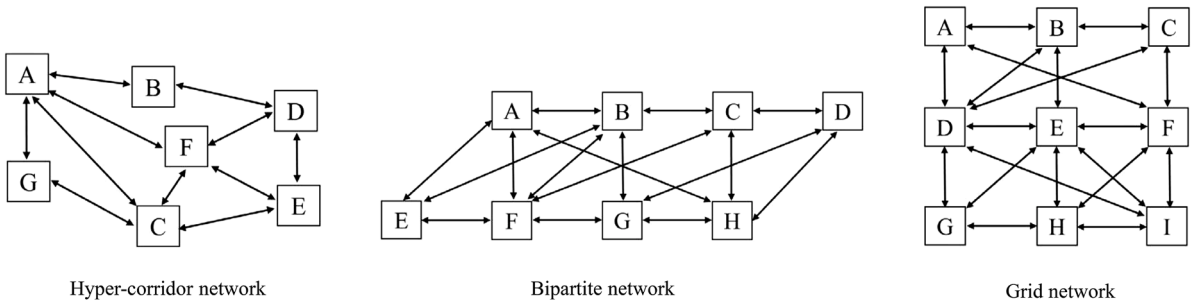


Fig. 4. Physical network topologies (Taherkhani et al., 2022).

Section 6.5 presents sensitivity analyses on the resilience of terminals in the SSNR problem, providing insights into the efficiency and adaptability of the SSNR model under different natural disaster scenarios.

6.1. Test instances

To closely align with real-world scenarios, three benchmarked physical network topologies are selected to simulate the physical network in its pre-disaster state (Taherkhani et al., 2022). These topologies are based on realistic service network configurations often encountered in North America, making them highly representative of scheduled service networks utilized in the logistics and transportation industries. As depicted in Fig. 4, the selected topologies include a hyper-corridor network, a bipartite network, and a grid network. Each topology offers a unique structural complexity, characterized by varying numbers of terminals and arcs, enabling comprehensive evaluation of the proposed MIP model across diverse disaster scenarios.

The hyper-corridor network represents a simplified yet practical structure with 7 terminals and 11 arcs, reflecting the linear, high-traffic routes commonly seen in major express service corridors. The bipartite network is moderately more complex, comprising 8 terminals and 16 arcs. Lastly, the grid network is the most intricate, featuring 9 terminals and 19 arcs, designed to represent dense urban or metropolitan service areas where demand is highly dispersed and interconnected.

These three network structures provide a balanced mix of complexity levels, making them ideal for evaluating the adaptability and performance of the proposed SSNR problem. To ensure a comprehensive evaluation, a total of 30 distinct test instances have been generated based on these real-world network topologies. These instances are sequentially numbered as P1 to P30. Specifically, instances P1 to P10 are derived from the hyper-corridor network, instances P11 to P20 are based on the bipartite network, and instances P21 to P30 utilize the grid network as their foundation.

It should be noted that the demands and the number of services are generated according to the physical network in the pre-disaster state. Each instance is designed to incorporate varying levels of complexity and disruption scenarios, enabling a robust assessment of the proposed MIP model under diverse uncertain conditions. The key attributes of these instances include the number of demands, the number of potential scheduled services, the time periods considered, and the specific terminals and arcs disrupted in the network. Detailed specifications for each instance are provided in Table 2.

Table 2
Details of the test instances.

Topology	Instances	Demands	Services	Time periods	Disrupted terminal	Disrupted arcs
		$\|\mathcal{K}\|$	$\|\Sigma\|$	$\|\mathcal{T}\|$		
Hyper-corridor network	P1	20	100	7	Terminal A	(D, E)
	P2	20	100	7	Terminal B	(C, E), (C,G)
	P3	20	100	7	Terminal C	(A, F)
	P4	30	100	14	Terminal D	(A, C), (C,E)
	P5	30	100	14	Terminal E	(A, G), (A, B)
	P6	30	100	14	Terminal F	-
	P7	30	100	14	Terminal G	(F, C), (F, E)
	P8	50	100	28	-	(A, G), (A, C), (A, F)
	P9	50	100	28	-	(F, C), (A, B), (A, D)
	P10	50	100	28	-	(B, D), (G, C), (A, F)
Bipartite network	P11	20	140	7	Terminal A	(E, F), (F, G)
	P12	20	140	7	Terminal B	(D, H), (H, G)
	P13	20	140	7	Terminal C	(A, B), (A, E)
	P14	40	140	14	Terminal D	(F, G), (G, H)
	P15	40	140	14	Terminal E	(B, C), (C, D)
	P16	40	140	14	Terminal F	(C, D), (D, H)
	P17	40	140	14	Terminal G	(A, E), (E, F)
	P18	60	140	28	Terminal H	(A, B), (B, C)
	P19	60	140	28	-	(A, B), (B, C), (C, D)
	P20	60	140	28	-	(E, F), (F, G), (G, H)
Grid network	P21	30	180	7	Terminal A	(D, E), (E, F)
	P22	30	180	7	Terminal B	(G, H), (H, I)
	P23	30	180	7	Terminal C	(A, D), (D, G)
	P24	50	180	14	Terminal D	(E, F)
	P25	50	180	14	Terminal E	-
	P26	50	180	14	Terminal F	(D, E)
	P27	50	180	14	Terminal G	(A, B), (B, C)
	P28	70	180	28	Terminal H	(I, E), (I, D)
	P29	70	180	28	Terminal I	(D, A), (A, B)
	P30	70	180	28	-	(A, B), (D, E), (E, F), (H, I)

The test instances summarized in Table 2 are designed to emulate realistic scenarios, using a structured random generation approach. The methodology is adapted from Taherkhani et al. (2022), ensuring both consistency with prior benchmarks and practical relevance. The essential characteristics of the instance generation process are detailed as follows:

- The demand volume for each instance, q_k, is determined via a uniform distribution, ranging between 10% and 50% of the service capacity, u_σ. Service capacities are also generated uniformly, spanning values from 40 to 200 units. The revenue associated with each demand, θ_k, is randomly generated within the range of 20 to 50 units.
- Each demand is assigned a randomly selected pair of terminals as the origin ($o_k \in \mathcal{N}^{\mathcal{PH}}$) and destination ($d_k \in \mathcal{N}^{\mathcal{PH}}$). The release time ($\alpha_k$) for demand pickup is generated uniformly between 1 and one-quarter of the total schedule length. Delivery time windows for each demand are defined by intervals starting from twice the release time (β_k^-) to the final schedule period (β_k^+).
- Delivery times are allowed to deviate by up to two periods before or after the specified time windows. A penalty cost of $\rho_k = 1$ per period is incurred for such deviations.
- For each pair of terminals, there is more than one parallel service. The duration of these services, τ_σ, is uniformly distributed over 1 to 3 time periods.
- The fixed cost for operating services is computed as $f_\sigma = \frac{300}{\tau_\sigma}$, reflecting the inverse relationship between duration and cost.
- Unit transportation costs for services ($c_{k\sigma}$) and unit holding costs at terminals (c_{kn}) are independently generated within a uniform range of 2 to 5.

6.2. Comparison of the BD reformulation with the exact solver

Here, we solve all the test instances using the BD reformulation algorithm in comparison with the exact solver. For the BD reformulation, the maximum number of iterations is set to 30. In this regard, Table 3 reports the solutions and CPU time required by the BD reformulation and exact solver in seconds.

Here, all test instances are solved using the BD reformulation algorithm and compared with the exact solver. For the BD reformulation, the maximum number of iterations is set to 30. Table 3 presents the objective values and CPU times (in seconds) obtained by both the BD reformulation and the exact solver. From the results, it is evident that the BD reformulation consistently finds solutions that are either equal to or very close to those obtained by the exact solver. For smaller instances (e.g., P1 to P10), the BD algorithm produces identical solutions with significantly lower computational times, demonstrating its efficiency. As the instance size increases, the exact solver requires considerably more time, whereas the BD reformulation maintains reasonable CPU times. For some larger

Table 3
Comparison of BD Reformulation with exact solver (CPU time is in seconds).

Instances	Exact Solver		BD Reformulation	
	Total profit	CPU Time	Total profit	CPU Time
P1	3745	30.71	3745	20.51
P2	3992	30.42	3992	20.12
P3	1120	32.71	1120	21.42
P4	11,208	52.76	11,208	22.81
P5	9980	59.71	9980	21.39
P6	11,852	56.38	11,852	21.33
P7	10,439	54.81	10,439	22.16
P8	29,342	35.82	29,342	24.81
P9	30,814	36.19	30,814	23.52
P10	28,742	35.74	28,742	21.63
P11	15,373	93.81	15,373	57.81
P12	18,619	87.52	17,688	54.28
P13	18,453	92.81	17,899	51.25
P14	39,221	234.81	36,083	58.91
P15	31,865	245.76	28,041	60.52
P16	33,159	226.81	28,848	63.17
P17	37,763	253.02	36,252	66.83
P18	59,168	875.32	59,168	69.81
P19	81,564	189.74	71,776	80.32
P20	79,327	175.92	76,154	71.58
P21	21,603	2351.84	20,523	351.62
P22	18,986	2261.47	16,708	383.51
P23	19,667	2411.83	17,504	418.62
P24	32,927	3182.65	28,976	422.59
P25	23,897	3027.51	19,596	416.38
P26	41,222	3104.73	36,688	381.43
P27	29,780	3267.91	24,717	525.82
P28	30,738	5723.81	26,127	519.73
P29	28,483	5688.25	26,204	523.91
P30	40,372	4561.84	38,757	528.76

instances (e.g., P12 to P30), slight deviations from the optimal objective value are observed. These deviations are primarily attributed to the premature termination criterion adopted for computational efficiency, whereby a fixed limit of 30 iterations is imposed for all large-scale instances. To address this issue, the BD reformulation is rerun using a revised termination criterion of an optimality gap less than or equal to 1% for instances in which an optimal solution is not obtained.

Fig. 5 illustrates the computational time comparison between the exact solver and the BD reformulation across all test instances. It is evident that the BD reformulation consistently requires significantly less CPU time than the exact solver, especially for larger instances (e.g., P14 to P30), where the exact solver exhibits a dramatic increase in computational effort. For smaller instances (i.e., P1 to P10), the BD approach still achieves faster solution times, highlighting its efficiency. Overall, the figure demonstrates that the BD reformulation provides a substantial reduction in computational time making it highly suitable for large-scale SSNR instances.

Fig. 6 illustrates the optimality gaps (%) obtained for the BD reformulation across problem instances P1 to P30. As shown, the BD reformulation achieves a zero optimality gap for a majority of instances (P1 to P11 and P18), indicating that it successfully identifies the exact optimal solutions in these cases. For the remaining instances, the optimality gaps are larger, ranging from approximately 3% to 18%, with the largest gaps observed in P25 and P27. These results suggest that while the BD approach is effective for small- to medium-sized instances, its performance decreases for certain large-scale or more complex instances.

For instances in which the BD reformulation did not achieve a zero optimality gap, the algorithm was rerun using a revised termination criterion of an optimality gap less than or equal to 1%, instead of the original stopping condition based on a maximum of 30 iterations. Fig. 7 shows that, although the CPU time of the BD under the revised termination condition is higher than that of the original version, it remains significantly lower than that of the exact solver and provides a reasonable computational time for obtaining optimal solutions to the SSNR model.

6.3. Performance metrics for disaster management

To evaluate the efficiency of the proposed MIP model, this paper analyzes its performance under two distinct operational states of the network: pre-disaster state and post-disaster state.

The pre-disaster state represents the physical network in a fully operational condition, without any disruptions. To model this state, constraints (10), (11), and (12) from the SSNR model in Section 4.2.2 are relaxed. In addition, constraint set (17) is replaced by constraint set (6) from the classical SSND formulation in Section 4.2.1, allowing the network to operate under normal conditions.

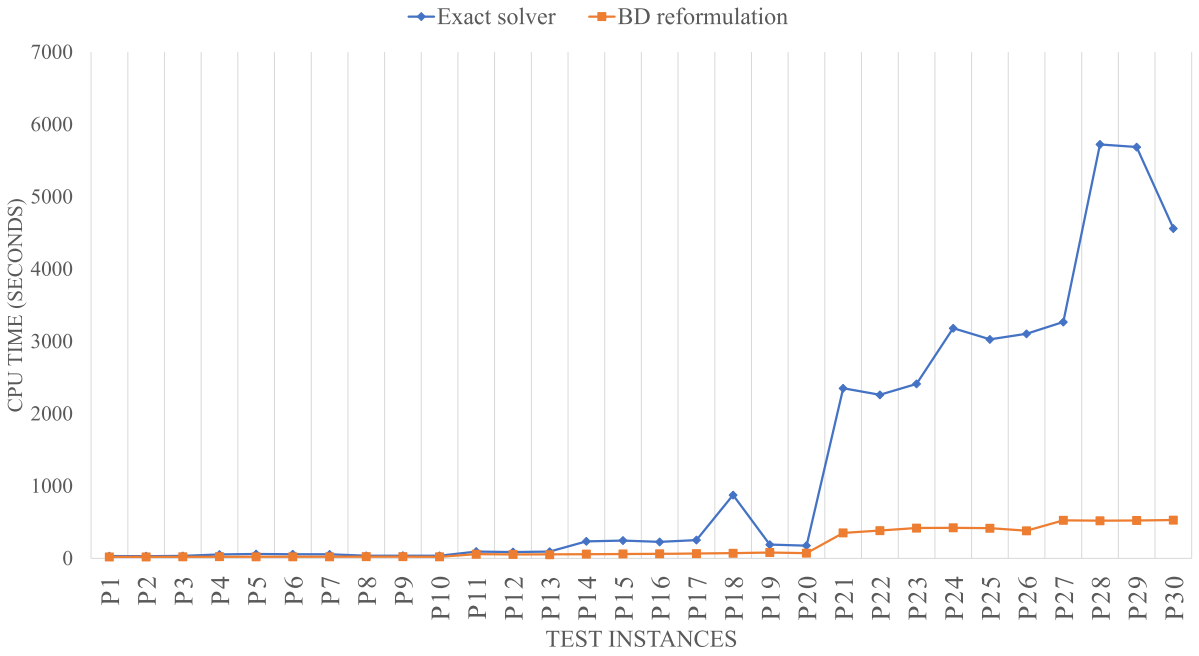


Fig. 5. Comparison based on the CPU time (in seconds).

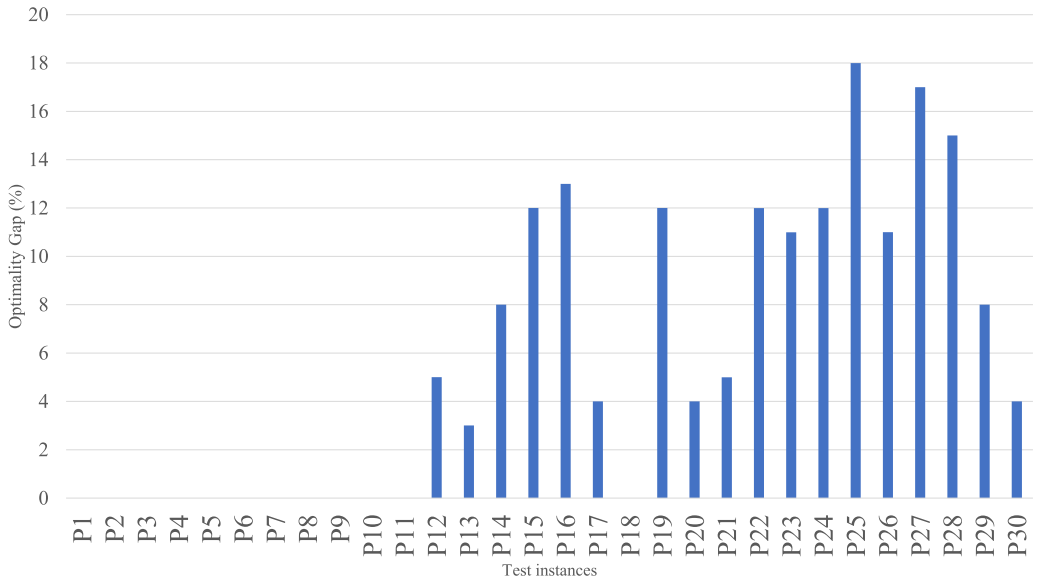


Fig. 6. Optimality gap for the BD reformulation.

The post-disaster state, on the other hand, captures the network under disruption. In this state, the full set of constraints from the MIP model presented in Section 4.2.2 is enforced, reflecting the reduced capacity and availability of terminals and services caused by the disaster.

To analyze the solutions for each network state, this study defines the following performance metrics:

- Total profit: this metric represents the value of the objective function, reflecting the overall profitability of the network operations.
- CPU time: this metric measures the computational time required for the CPLEX solver to find an optimal solution.
- Demands: this metric indicates the total number of demands accepted by the SSNR model.
- Services: this metric refers to the total number of active scheduled services utilized in the physical network.
- Usage (%): this metric represents the average capacity utilization across all active services.

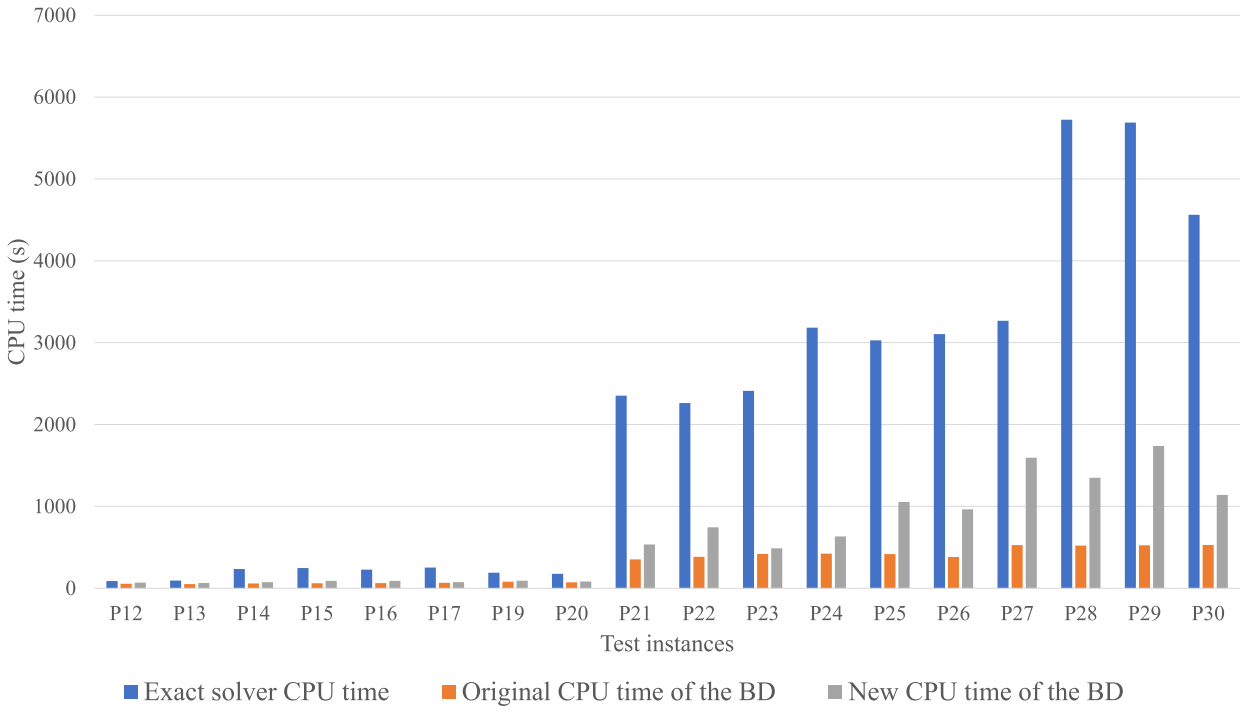


Fig. 7. Comparison of CPU time (in seconds) across test instances for the exact solver and two versions of the BD algorithm.

These metrics enable a comprehensive evaluation of the SSNR model's performance under both pre-disaster and post-disaster conditions.

6.4. Analyzing the pre-disaster and post-disaster scenarios

For each network topology, this paper reports the results of the test instances based on the performance metrics under both pre- and post-disaster conditions. The analyses corresponding to the hyper-corridor network are presented in Table 4, whereas the analyses for the bipartite network are provided in Table 5. Furthermore, the analyses for the grid network are summarized in Table 6. Finally, the tradeoff between optimality and computational complexity across all instances is illustrated in Fig. 8.

The computational results presented in Table 4 offer valuable insights into the performance metrics of the hyper-corridor network under both pre-disaster and post-disaster states. In the pre-disaster state, the network design exhibits consistently high performance. For smaller instances (e.g., P1 to P3), the solution values remain stable at 5584 with an average CPU time of approximately 23 seconds. These instances accept 20 demands and open 19 services, resulting in a capacity usage of 73.62%. For larger instances, such as P8 to P10, the solution value increases significantly to 26027, and the capacity usage rises to 83.71%, demonstrating the network's ability to handle higher demand efficiently. Notably, the CPU time for these larger instances also increases, reflecting the additional computational effort required for optimization.

In the post-disaster state, the network's performance declines due to the disruptive effects on services and demand fulfillment. For example, in instance P1, the solution value drops from 5584 to 3745, and the number of accepted demands decreases from 20 to 16. Despite the increase in open services from 19 to 22, the capacity usage drops significantly to 50.86%. This pattern is consistent across other instances, indicating that post-disaster conditions lead to reduced network efficiency and resource utilization. For larger instances, e.g., P8 to P10, the network redesign process demonstrates a more robust performance in the post-disaster state. While the solution values decrease compared to the pre-disaster state, they remain relatively high, ranging from 28,742 to 30814. Similarly, the capacity usage for these instances stays above 70%, suggesting that the network retains considerable capability despite the disruptions. The CPU time for these instances in the post-disaster state is slightly higher than in the pre-disaster state, indicating the added complexity of optimizing under constrained conditions.

The computational results presented in Table 5 highlight the performance of the bipartite network topology across various instances under pre- and post-disaster states. Overall, the network redesign process demonstrates consistent solution quality across all instances, with notable differences in resource usage, service allocation, and computational time between the two states. The pre-disaster scenarios exhibit lower computational demands, higher resource efficiency, and a more balanced allocation of services compared to the post-disaster state, where the network faces heightened challenges in meeting demands effectively.

The SSNR model achieves efficient solutions in the pre-disaster state with relatively low CPU times and efficient resource utilization. For instance, addressing the test P11 utilizes 64.92% of the available capacity to meet 20 demands with 14 services, and the

Table 4
Computational results for the hyper-corridor network (CPU time is in seconds).

Instances	Pre-disaster state					Post-disaster state				
	Total profit	CPU time	Demands	Services	Usage (%)	Total profit	CPU time	Demands	Services	Usage (%)
P1	5584	23.81	20	19	73.62	3745	30.71	16	22	50.86
P2	5584	23.91	20	19	73.62	3992	30.42	17	24	47.28
P3	5584	23.12	20	19	73.62	1120	32.71	14	19	65.41
P4	17296	25.61	30	24	72.53	11208	52.76	24	28	56.24
P5	17296	25.11	30	24	72.53	9980	59.71	20	30	34.85
P6	17296	25.98	30	24	72.53	11852	56.38	21	24	43.92
P7	17296	24.81	30	24	72.53	10439	54.81	23	27	39.85
P8	26027	32.91	50	32	83.71	29342	35.82	50	39	73.92
P9	26027	32.91	50	32	83.71	30814	36.19	50	40	70.28
P10	26027	32.91	50	32	83.71	28742	35.74	50	37	75.27

Table 5
Computational results for the bipartite network (CPU time is in seconds).

Instances	Pre-disaster state					Post-disaster state				
	Total profit	CPU time	Demands	Services	Usage (%)	Total profit	CPU time	Demands	Services	Usage (%)
P11	26742	66.12	20	14	64.92	15373	93.81	12	16	38.91
P12	26742	67.92	20	14	64.92	18619	87.52	16	15	46.71
P13	26742	65.35	20	14	64.92	18453	92.81	17	13	52.54
P14	46134	79.81	40	31	54.81	39221	234.81	32	38	32.81
P15	46134	82.54	40	31	54.81	31865	245.76	31	39	45.27
P16	46134	76.92	40	31	54.81	33159	226.81	31	35	42.91
P17	46134	77.05	40	31	54.81	37763	253.02	30	41	37.81
P18	86305	154.82	60	45	64.56	59168	875.32	44	52	35.75
P19	86305	141.09	60	45	64.56	81564	189.74	60	55	53.91
P20	86305	147.32	60	45	64.56	79327	175.92	60	62	41.58

Table 6
Computational results for the grid network (CPU time is in seconds).

Instances	Pre-disaster state					Post-disaster state				
	Total profit	CPU time	Demands	Services	Usage (%)	Total profit	CPU time	Demands	Services	Usage (%)
P21	28369	1898.32	30	54	62.76	21603	2351.84	25	61	58.87
P22	28369	1763.27	30	54	62.76	18986	2261.47	23	57	60.51
P23	28369	1842.25	30	54	62.76	19667	2411.83	24	55	57.81
P24	43709	2419.74	50	78	73.44	32927	3182.65	45	68	75.71
P25	43709	2305.82	50	78	73.44	23897	3027.51	38	72	54.28
P26	43709	2461.94	50	78	73.44	41222	3104.73	42	76	78.97
P27	43709	2503.61	50	78	73.44	29780	3267.91	39	81	53.84
P28	42142	4216.93	70	105	76.18	30738	5723.81	58	121	48.91
P29	42142	4527.81	70	105	76.18	28483	5688.25	59	122	47.65
P30	42142	4362.25	70	105	76.18	40372	4561.84	70	109	72.81

computational time remains moderate at 66.12 seconds. Similarly, larger instances such as P18 demonstrate scalability, achieving a solution value of 86,305 with 64.56% resource utilization and 154.82 seconds of CPU time. These results underscore the SSNR model's capacity to efficiently manage network operations in pre-disaster conditions, ensuring that resources are optimally deployed to meet service demands.

The post-disaster state, by contrast, reveals the SSNR model's adaptability under disrupted conditions but also highlights increased resource strain and computational complexity as the network redesign is more complex than the network design. For example, in test P11, resource utilization drops significantly to 38.91%, and computational time rises to 93.81 seconds as the network adjusts to reduced demand and service reallocation. Similar trends are observed in larger instances like P18, where resource usage decreases to 35.75%, and computational time escalates to 875.32 seconds. These findings reflect the challenges of maintaining service efficiency and resource allocation during disruptions, emphasizing the importance of the optimal network redesign process for the scheduled services.

Table 6 presents a comparative analysis of computational results for the grid network in both pre-disaster and post-disaster states. In general, the pre-disaster state in the grid network consistently achieves higher solution values and lower CPU times compared to the post-disaster state. This indicates that SSND is easier than the SSNR in terms of computational effort and outcomes under normal conditions. However, the post-disaster state reveals the complexity introduced by disruptions, leading to a decline in solution values and increased computational times. The average capacity usage also varies notably between the two states, reflecting changes in network configurations and adjustments required to handle disruptions.

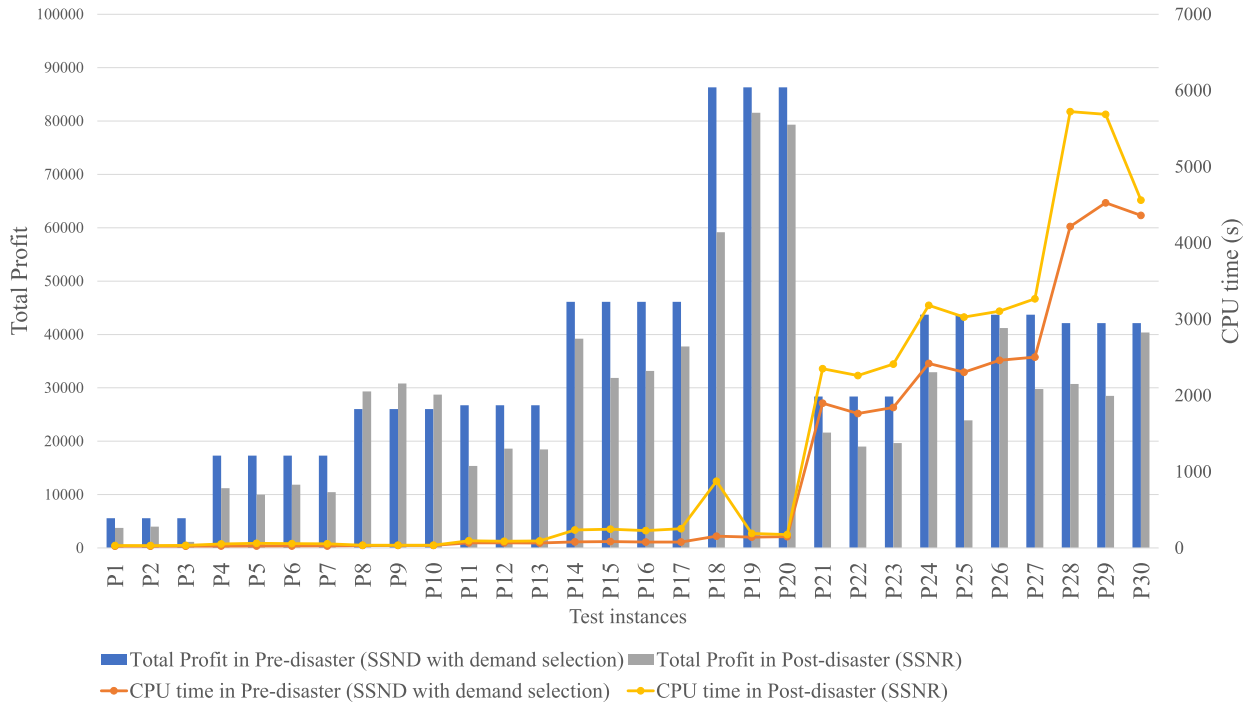


Fig. 8. Tradeoff between optimality and complexity criteria in the classical SSND with demand selection (Pre-disaster state) and the proposed SSNR (Post-disaster state).

In the pre-disaster state, instances such as P21 to P23 accept 30 demands with 54 open services, achieving an average capacity usage of 62.76%. In contrast, the post-disaster state sees a reduction in accepted demands (e.g., 25 for P21, 23 for P22, and 24 for P23) and varying numbers of open services (e.g., 61, 57, and 55, respectively). This reduction leads to lower average usage rates of 58.87%, 60.51%, and 57.81%, respectively. These results underscore the challenges in maintaining high network performance post-disaster, as fewer demands can be accommodated despite the increased number of open services. For instances with higher pre-disaster demands and services (e.g., P24–P27, with 50 demands and 78 open services), the pre-disaster average usage is 73.44%. Post-disaster adjustments result in fewer accepted demands (e.g., 45 for P24 and 39 for P27) and significant changes in the number of open services (e.g., 68 for P24 and 81 for P27). Interestingly, the post-disaster average usage varies, with some instances (e.g., P24) exhibiting an increase to 75.71%, while others (e.g., P27) show a decline to 53.84%. This variability reflects the trade-offs between service availability and demand coverage in disrupted scenarios. In cases with the largest demands and services (e.g., P28 to P30, with 70 demands and 105 open services), the pre-disaster average usage reaches 76.18%. Post-disaster, the accepted demands range from 58 to 70, while open services increase to 121–122. The post-disaster average usage exhibits notable declines (e.g., 48.91% for P28 and 47.65% for P29), except for P30, which maintains a high usage rate of 72.81%.

In the computational results for the hyper-corridor network, bipartite network, and grid network, instances with only a few disrupted arcs with no terminal disruptions, such as P8, P9, P10, P19, P20, and P30, show no unaccepted demands. Conversely, when a terminal is disrupted, the SSNR model is forced to reject a few demands, underscoring the severe impact of terminal disruptions compared to arc disruptions. These findings emphasize the critical importance of adaptive network design strategies to maintain high utilization rates under real-world disruptions. The results highlight the necessity for efficient redesign strategies in the post-disaster state, where satisfying all demands is not feasible. While increasing the number of open services can partially mitigate demand loss, achieving high utilization and maintaining computational efficiency remain significant challenges for the SSNR problem.

Finally, Fig. 8 further illustrates the tradeoff between solution quality and computational complexity for the pre-disaster state, represented by the classical SSND with demand selection, and the post-disaster state, represented by the proposed SSNR model. Although both models employ the same objective structure and demand selection mechanism, the SSNR consistently requires higher CPU times across most test instances due to the disruption-driven feasibility interactions introduced by the post-disaster network structure. In particular, the post-disaster environment forces the SSNR model to simultaneously evaluate service availability, disrupted terminal accessibility, alternative itinerary construction, and the economic viability of serving each demand under restricted network conditions.

Fig. 8 also shows that the post-disaster solutions generally yield lower total profits compared with the pre-disaster SSND solutions, reflecting the loss of feasible services and the increased operational costs associated with rerouting and reduced network connectivity after disruptions. More importantly, the computational gap between the SSND and SSNR becomes increasingly significant for medium- and large-scale instances, indicating that the disrupted-network setting substantially enlarges the combinatorial complexity of the

Table 7
Sensitivity analyses on the terminals in the hyper-corridor network (CPU time is in seconds).

Terminals	Total profit	CPU time	Demands	Services	Usage (%)
A	3815	29.84	16	20	54.82
B	4086	28.19	17	21	61.29
C	1220	33.81	14	18	74.71
D	2346	32.71	14	19	68.42
E	3291	34.84	16	20	56.37
F	1552	37.81	13	17	65.83
G	3275	30.52	16	18	72.55

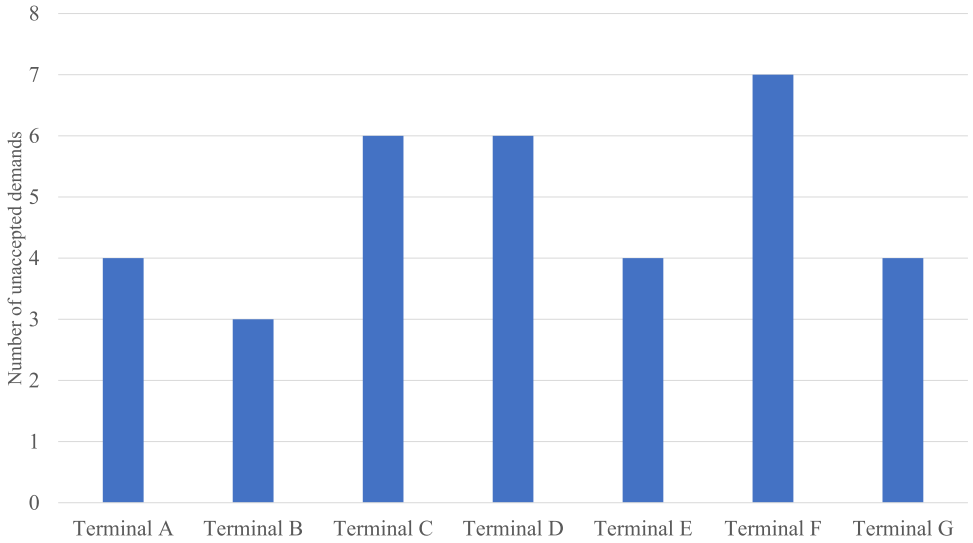


Fig. 9. Impact of disrupted terminals on the unsatisfied demands.

decision space. Therefore, the SSNR cannot be interpreted merely as a standard SSND solved on a preprocessed network with removed arcs or terminals. Instead, the disruptions fundamentally modify the interaction between demand selection, service activation, and itinerary feasibility decisions within the optimization process.

6.5. Sensitivity analyses on the network resilience

This section presents a detailed sensitivity analysis to evaluate the robustness and efficiency of the proposed SSNR model in assessing network resilience. In particular, the effects of network redesign are investigated. Here, this study focused on a hyper-corridor network consisting of 20 demands, $|\mathcal{K}| = 20$, 100 potential scheduled services, $|\Sigma| = 100$, and 7 time periods, $|T| = 7$. Table 7 presents the results of the network redesign when disruptions occur at individual terminals within the hyper-corridor network. Based on the data in Table 7 and its comparison with a pre-disaster state, Fig. 9 illustrates the number of unaccepted demands for each disrupted terminal, while Fig. 10 provides a comparison of the computational time complexity for each disruption scenario.

Table 7 presents the results of sensitivity analysis for disruptions at terminals in the hyper-corridor network, highlighting their impact on solution quality, computational time, and overall network performance. Terminal B achieves the highest total profit value, i.e., 4086, with the lowest computational time, i.e., 28.19 seconds, indicating its efficiency and robustness under disruption scenarios. Conversely, Terminal C reports the lowest total profit, i.e., 1220, despite its high usage efficiency of 74.71%, suggesting a limited overall contribution to network profitability. Terminals A, E, and G manage the highest number of demands, i.e., 16 each, demonstrating their critical role in sustaining network functionality. Meanwhile, Terminal F shows the lowest service usage, 17 services, and the highest computational time, 37.81 seconds, indicating potential inefficiencies.

As shown in Fig. 9, Terminal F plays a key role in generating the highest number of unsatisfied demands under disruption, highlighting its critical influence on network reliability. In contrast, disruptions at Terminal B have the least impact on increasing the number of unaccepted demands, underscoring its robustness. Furthermore, Fig. 10 illustrates that disruptions at Terminal F result in the highest computational complexity. Conversely, disruptions at Terminal B exhibit the lowest computational complexity among all cases, reinforcing its stability and operational efficiency.

The sensitivity analysis of individual terminals provides quantitative insights into their operational resilience under disruption scenarios. In this study, resilience is assessed primarily by the number of accepted demands, which reflects a terminal's ability to sustain service continuity when disruptions occur. Terminal B demonstrates the highest resilience, handling 17 demands and

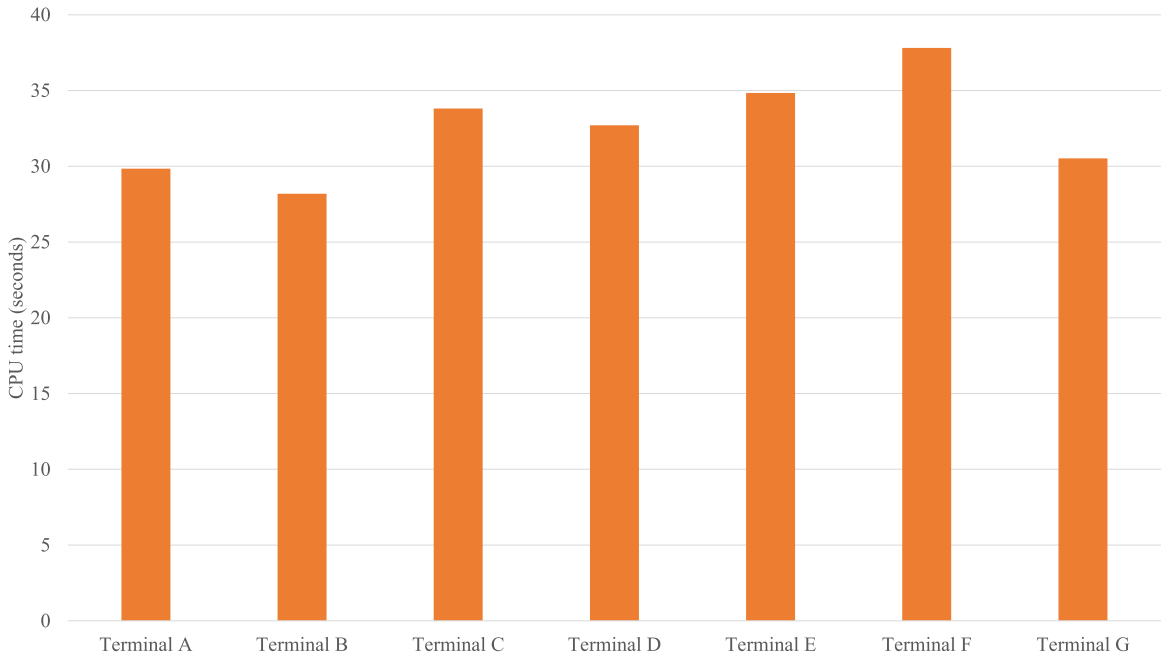


Fig. 10. Impact of disrupted terminals on the computational complexity.

contributing the highest solution value, indicating that it plays a critical role in maintaining overall network performance. Terminals A, E, and G each accommodate 16 demands with moderately high solution values, reflecting strong resilience and highlighting their importance in supporting network functionality, although slightly less critical than Terminal B.Conversely, Terminal F shows the lowest resilience, with only 13 accepted demands and a low solution value, suggesting that it is particularly vulnerable to disruptions and may require strategic interventions such as capacity reinforcement or operational improvements. Terminals C and D exhibit moderate resilience, handling 14 demands each, but their higher service usage percentages suggest they may become bottlenecks under additional stress. These quantitative insights provide actionable guidance for transportation network planners, emphasizing targeted investments in highly resilient terminals to preserve service continuity and prioritizing interventions for weaker terminals to enhance overall network robustness.

7. Conclusions and perspectives

This paper introduces the SSNR problem, an extension of the classic SSND problem, aimed at optimizing transportation service networks in the aftermath of natural disasters. The proposed MIP model addresses network disruptions by maximizing total profit, selecting demands and available services, and generating demand itineraries. To manage the complexity arising from the simultaneous selection of demands and services in a disrupted network, a BD reformulation is applied, and its performance is compared against the exact solver. Furthermore, computational experiments conducted on hyper-corridor, bipartite, and grid network topologies provide valuable insights into the behavior and effectiveness of the SSNR model under both pre- and post-disaster conditions.

The results indicate that while classical SSND models perform efficiently under pre-disaster conditions, post-disaster scenarios pose substantial challenges due to disruptions at terminals and along arcs. The proposed SSNR model demonstrates its practical value by enabling targeted recovery strategies and optimized resource allocation under predetermined disruption scenarios. Analysis shows that terminal disruptions lead to more pronounced reductions in service coverage and overall network efficiency compared to arc disruptions. Specifically, disruptions at a single high-traffic terminal can substantially increase total operational costs, highlighting the critical nodes that should be prioritized for reinforcement. Moreover, the SSNR model identifies routes and services that can be optimally reassigned to maintain service levels for high-priority demands, thereby reducing overall system downtime. These findings provide actionable guidance for transportation network planners by indicating the optimal combination of terminal investments and flexible service reassignment strategies to enhance system resilience, including recommendations on which terminals to reinforce first and which services to prioritize during post-disaster recovery.

A limitation of the present study lies in the generation of disruption scenarios. While the deterministic selection of disrupted nodes follows standard SSND benchmarks and allows for consistent computational experiments, it does not account for spatial correlations inherent in real-world disasters. Natural disasters such as floods or earthquakes typically impact contiguous areas, affecting multiple interconnected facilities simultaneously. Consequently, the current approach may underestimate network vulnerability and the true impact of regional failures. Future work should consider integrating spatially correlated or region-based disruption models to more accurately evaluate supply chain resilience under realistic disaster scenarios.

Another limitation of the current study is the assumption of fixed terminal capacities in the post-disaster scenario, which does not fully capture the rapidly changing conditions or the potential need for temporary capacity adjustments. In real-world disaster situations, terminals may need to adapt quickly to fluctuating demand and resource availability, necessitating a more dynamic approach to capacity management. Future research could extend the SSNR model by incorporating temporary capacity configurations at terminals to better reflect these operational challenges and improve the SSNR model's applicability in real-world disaster response planning.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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